

C3 Network

Hands-on with Filius – a network simulator

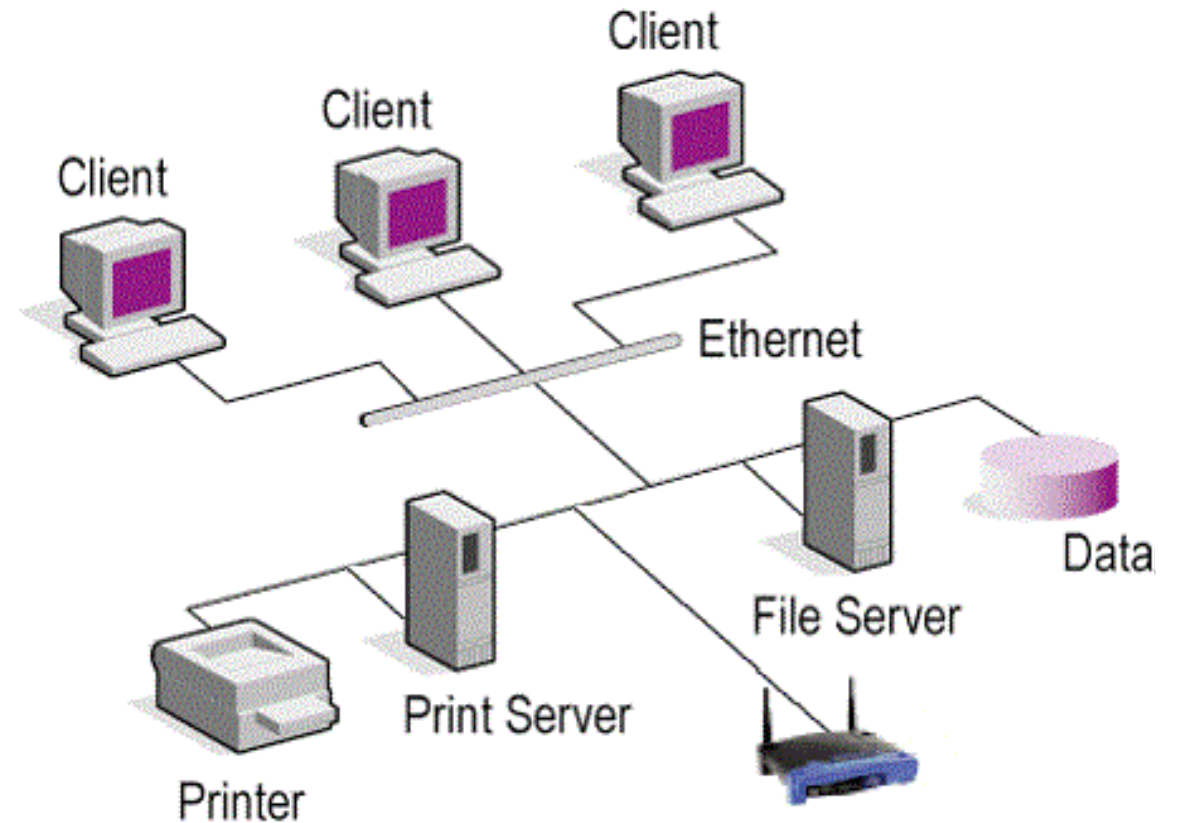
Part 1a : Network Infrastructure

Hosts, NIC, Medium

Hosts and Nodes

What are Hosts ?

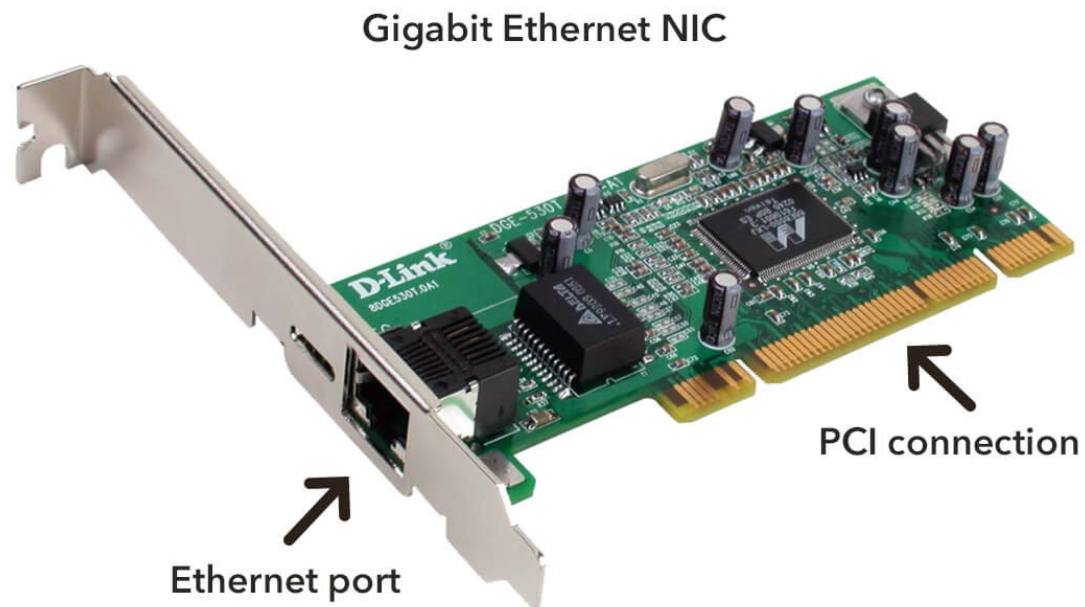
- Hosts are servers and clients



What are Nodes ?

- Every network device must have at least one node
- Examples : Network Interface Card (NIC),
Wifi, Bluetooth or Infrared Ports

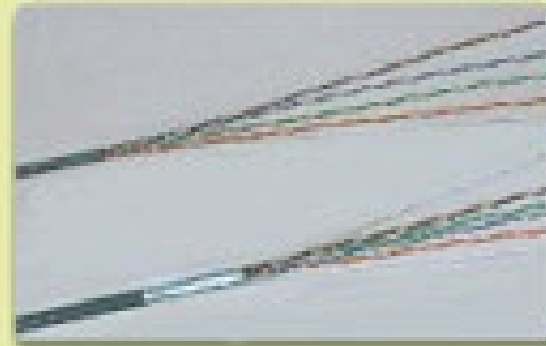
Network Interface Card (NIC)



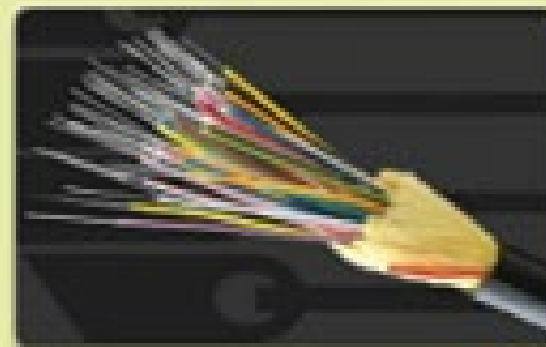
Wifi interface card

Network Medium Types

Copper



Fiber Optic



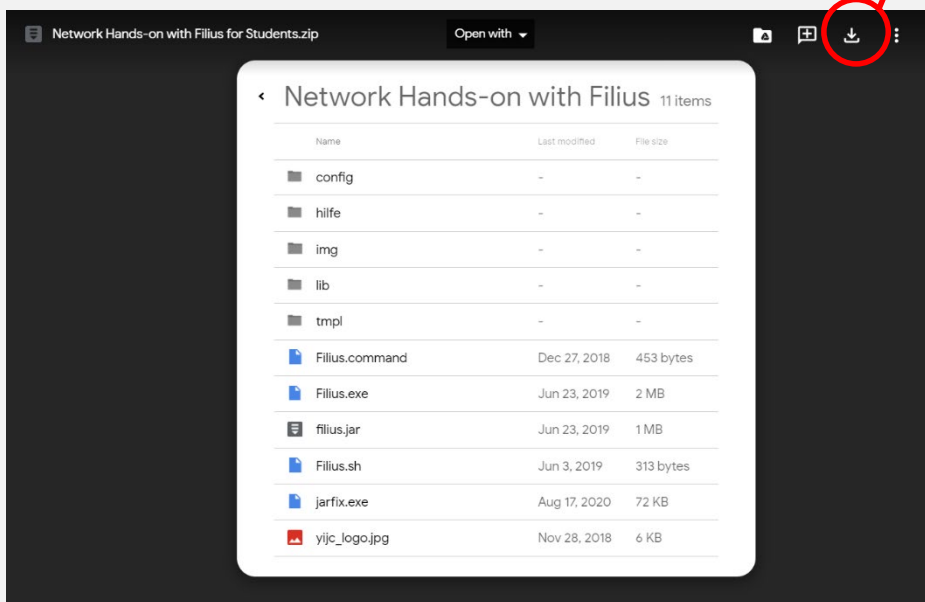
Wireless



Installation and Introduction to Filius

Installation

1. Download the Hands-on zip file either from Google Drive or Coursemology.



Click here to download (in Google Drive)

2. Launch of the program either by double clicking on

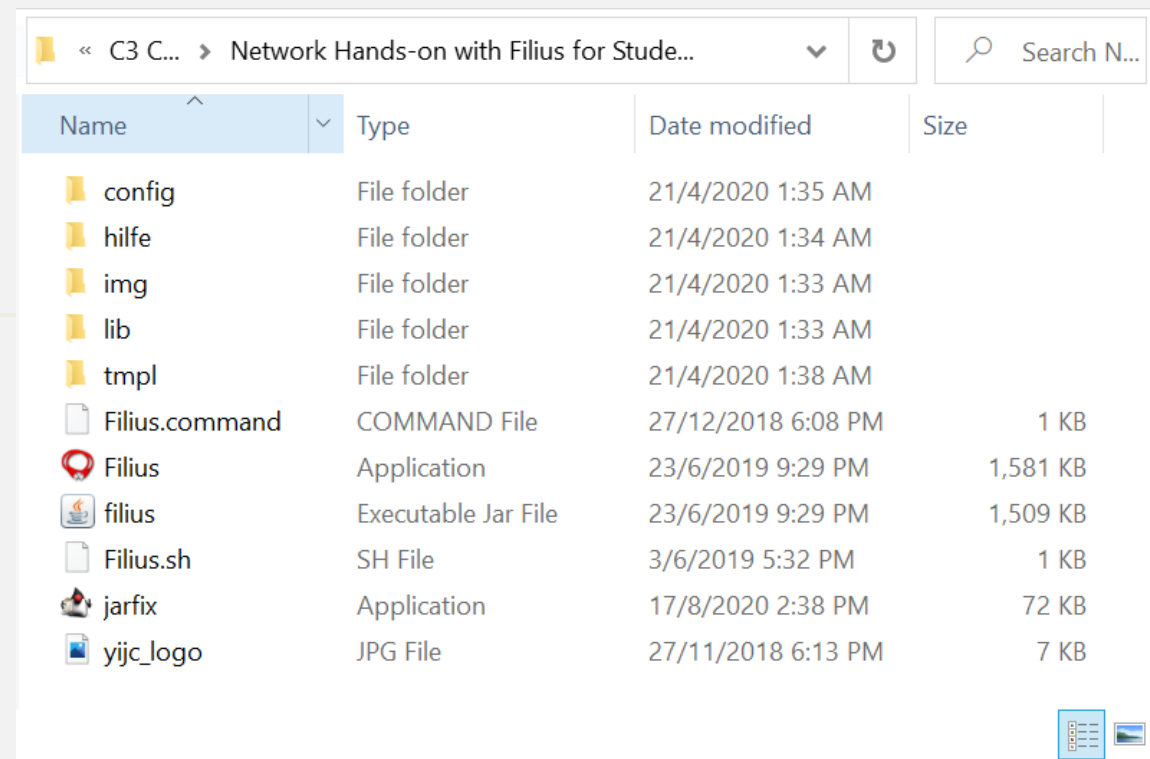
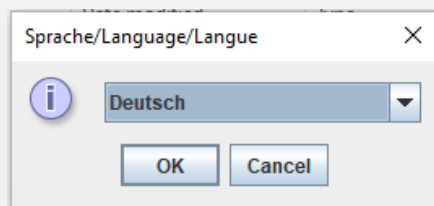


filius.exe or



filius.jar

3. Select the English language to use.

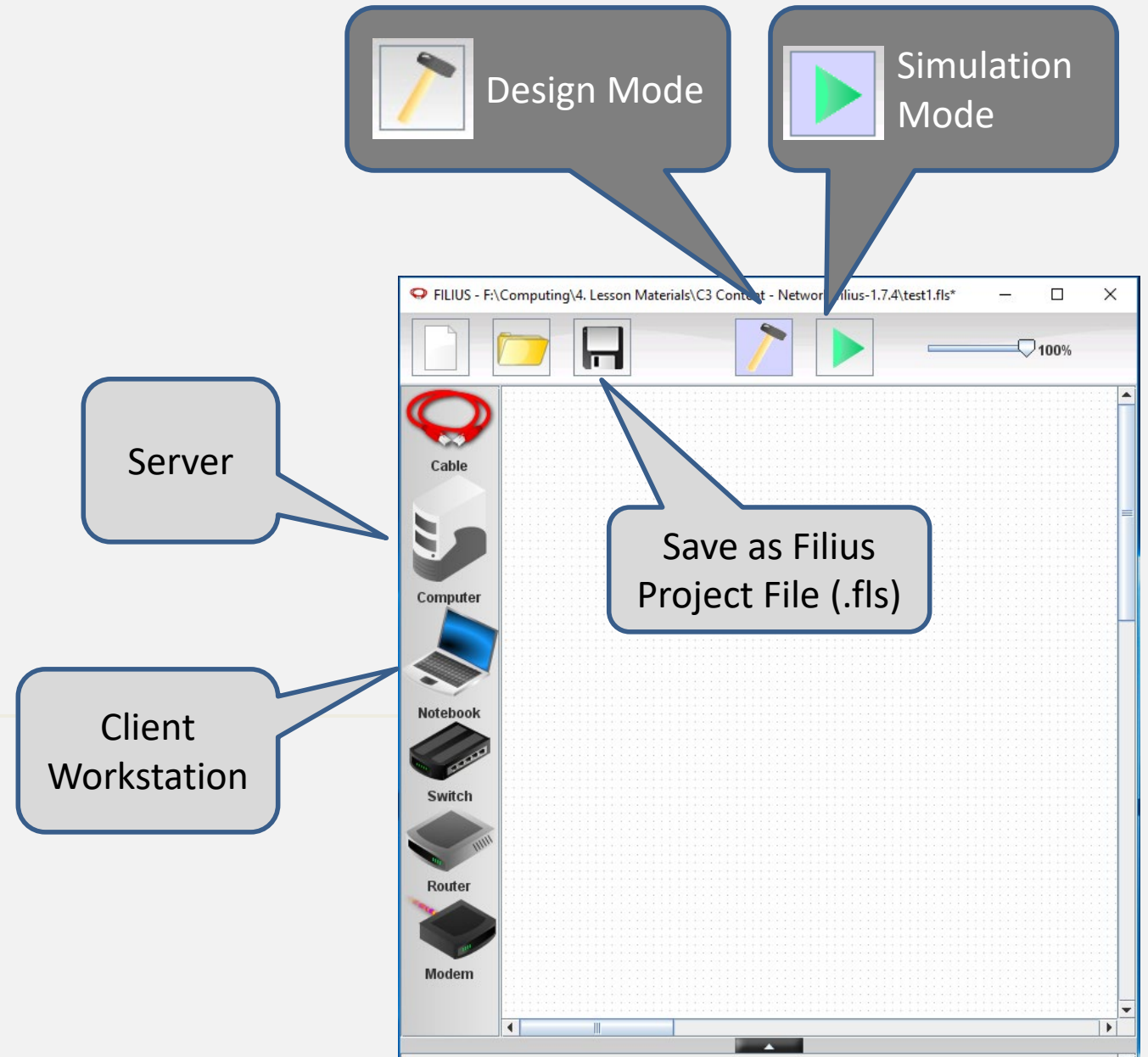


Introduction to Filius

FILIUS was developed by the University Siegen, Germany, for enhancing the teaching of computer networks.

Drag-and-drop the components to workspace.  to deselect.

 to remove. *Double-click* (or *Right-click*) to configure the component.



ICONS



Important



For curious mind

Hands-on 1:

Construct a Peer-to-Peer Network

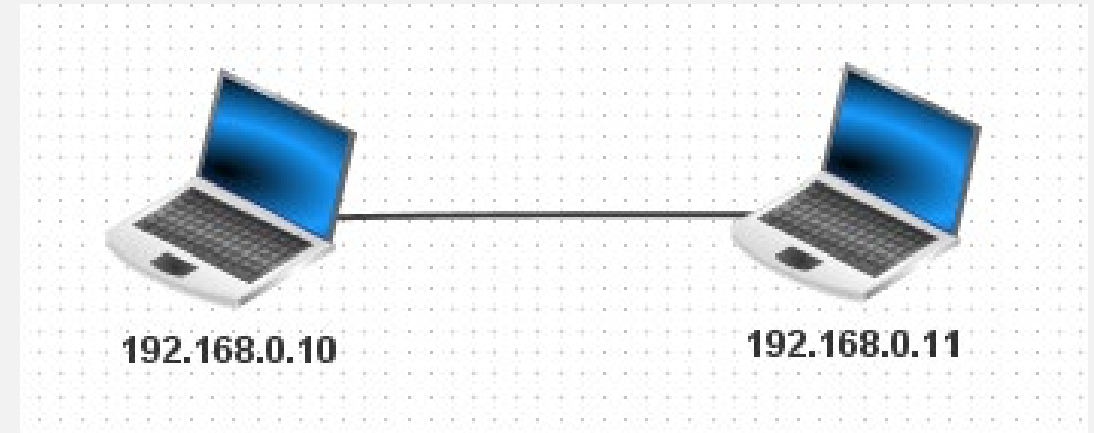
Construct a Peer-to-Peer Network



Create a simple network with two linked Notebook.

Configure the IP addresses of the Notebooks to 192.168.0.10 and 192.168.0.11.

Using the subnet mask 255.255.255.0 ensures that both computers are part of the same network.



Name	192.168.0.10	☒ Use IP address as Name
MAC Address	0F:7B:CC:D1:BB:11	☐ Use DHCP for configuration
IP address	192.168.0.10	DHCP server setup
Netmask	255.255.255.0	
Gateway		
Domain Name Server		



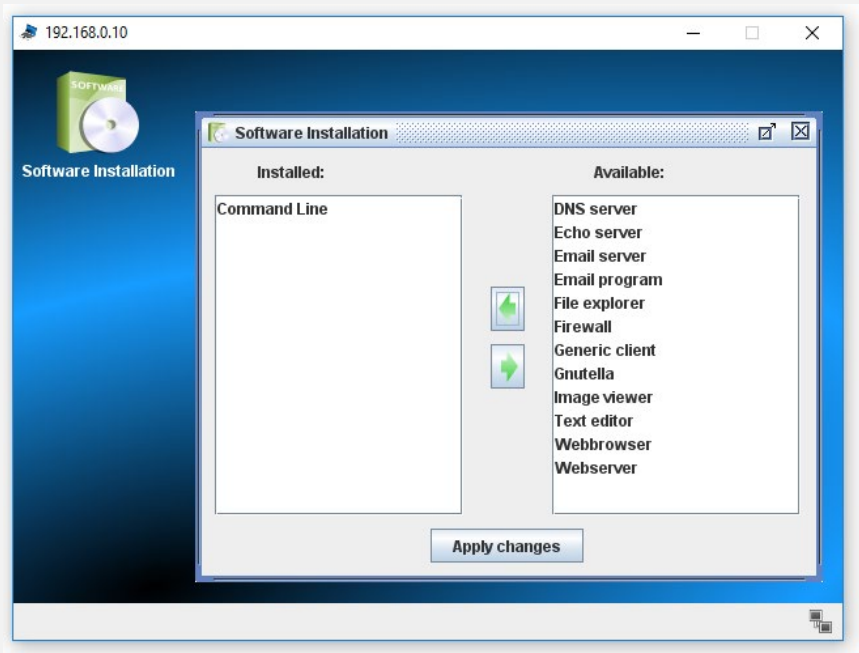
A crossover cable is used.

<https://community.fs.com/blog/patch-cable-vs-crossover-cable.html>

Construct a Peer-to-Peer Network

[Simulation Mode ]

Click and select the Notebook with IP address 192.168.0.10 to install the Command Line on this Notebook.



Circuit Switching vs **Packet Switching**?

<https://www.youtube.com/watch?v=B1tElYnFqL8>

```

Command Line

=====
List of available commands:
arp          show Address Resolution Protocol (ARP) table
cat / type   show file contents
cd           change directory
copy / cp    copy file
del / rm     delete file/directory
dir / ls     show list of files in current directory
exit         exit terminal application
help        show this list of commands
host         resolve hostname to IP address
ipconfig     show network configuration
mkdir       create directory
move / mv    move/rename file
netstat      show list of connections
ping        test connection to other computer
pwd         print working directory
route       show routing table
touch       create file
tracert      analyse hops of connecting path
=====

root /> ping 192.168.0.11
  
```

Type Ping 192.168.0.11 to test the connection with the other Notebook.

Try-It : ipconfig

```

root /> ping 192.168.0.11
PING 192.168.0.11 (192.168.0.11)
From 192.168.0.11 (192.168.0.11): icmp_seq=1 ttl=64 time=217ms
From 192.168.0.11 (192.168.0.11): icmp_seq=2 ttl=64 time=102ms
From 192.168.0.11 (192.168.0.11): icmp_seq=3 ttl=64 time=104ms
From 192.168.0.11 (192.168.0.11): icmp_seq=4 ttl=64 time=103ms
--- 192.168.0.11 packet statistics ---
4 packet(s) transmitted, 4 packet(s) received, 0% packet loss
  
```

Success!
All 4 **packets** are transmitted and received.



What is a **packet**?

<https://computer.howstuffworks.com/question525.htm>

Try-It : Using a switch for Peer-to-Peer Network

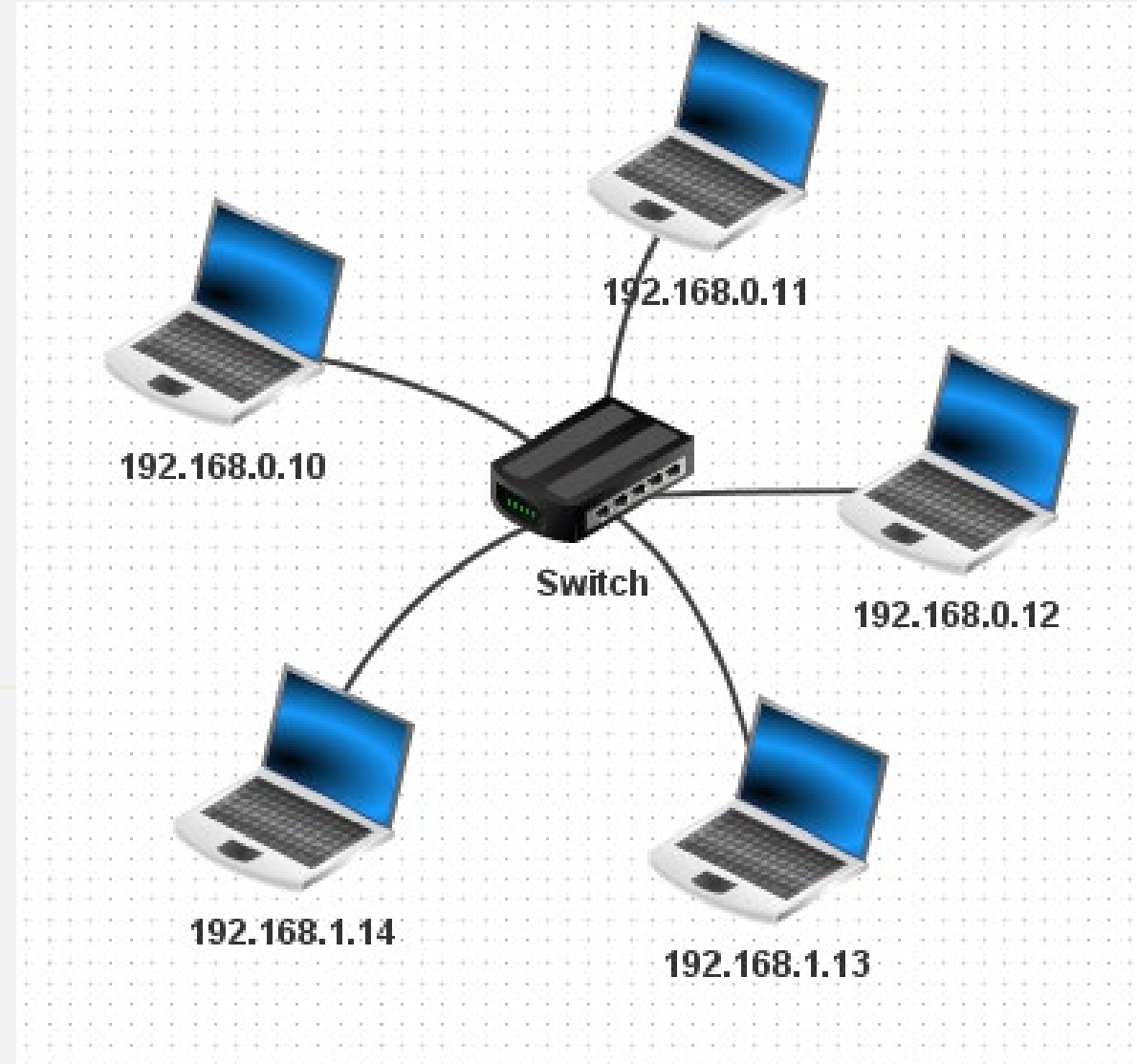


Use a switch to connect some Notebooks together.

Check the connections using `ping` in the Command Line.

Note that 192.168.1.13 and 192.168.1.14 are located on a separate network.

Try to ping each other from different notebooks.



Secret of Subnet Mask

<https://www.youtube.com/watch?v=yLeuGOOrUvo>

Peer-to-Peer Network

Source:

https://www.teach-ict.com/gcse_new/networks/peer_peer/miniweb/pg5.htm

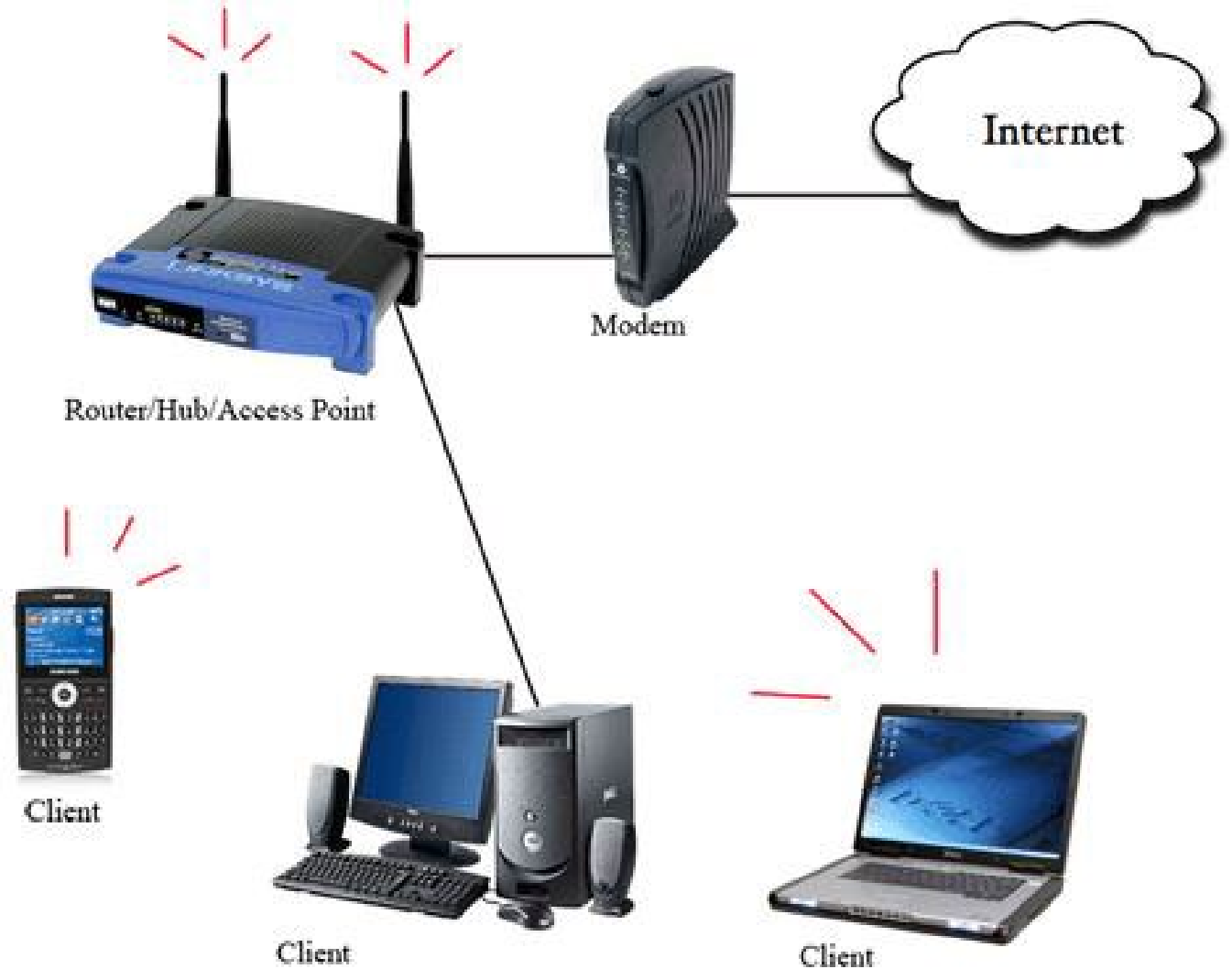
Advantages	Disadvantages
No need for a network operating system	Because each computer might be being accessed by others it can slow down the performance for the user
Does not need an expensive server because individual workstations are used to access the files	Files and folders cannot be centrally backed up
No need for specialist staff such as network technicians because each user sets their own permissions as to which files they are willing to share.	Files and resources are not centrally organised into a specific 'shared area'. They are stored on individual computers and might be difficult to locate if the computer's owner doesn't have a logical filing system.
Much easier to set up than a client-server network - does not need specialist knowledge	Ensuring that viruses are not introduced to the network is the responsibility of each individual user
If one computer fails it will not disrupt any other part of the network. It just means that those files aren't available to other users at that time.	There is little or no security besides the permissions. Users often don't need to log onto their workstations.

Part 1a : Network Infrastructure

Network Types

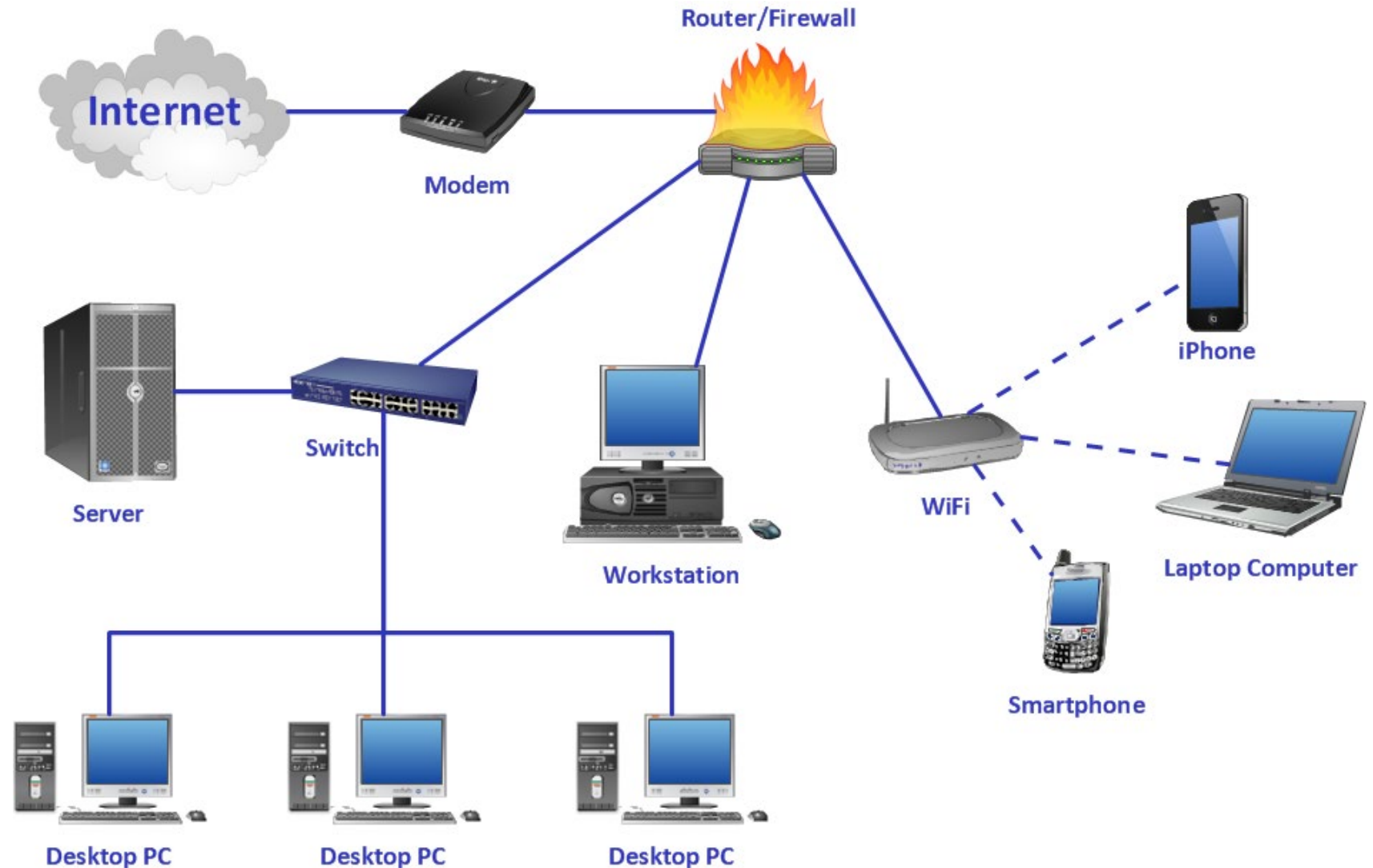
Networking

- The Simplest: Peer-to-Peer Network
- Most familiar: Personal Area Network (PAN or Home Network)



Local Area Network (LAN)

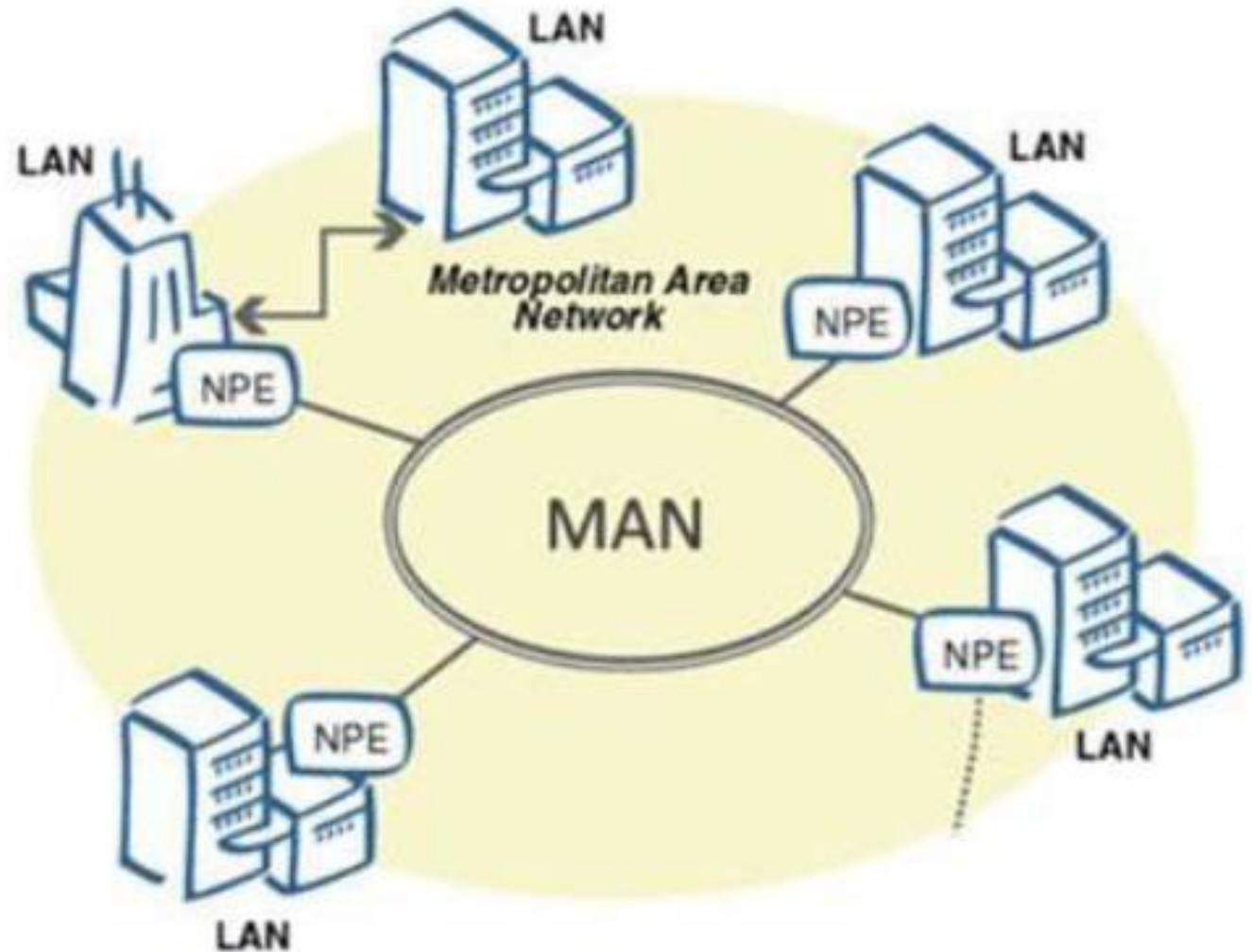
A local area network (LAN) need include equipment for internet access.



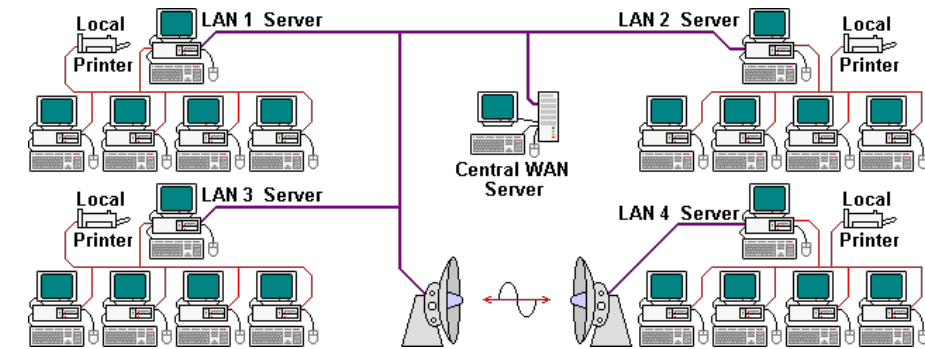
Metropolitan Area Network (MAN)

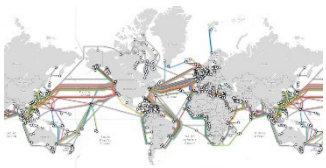
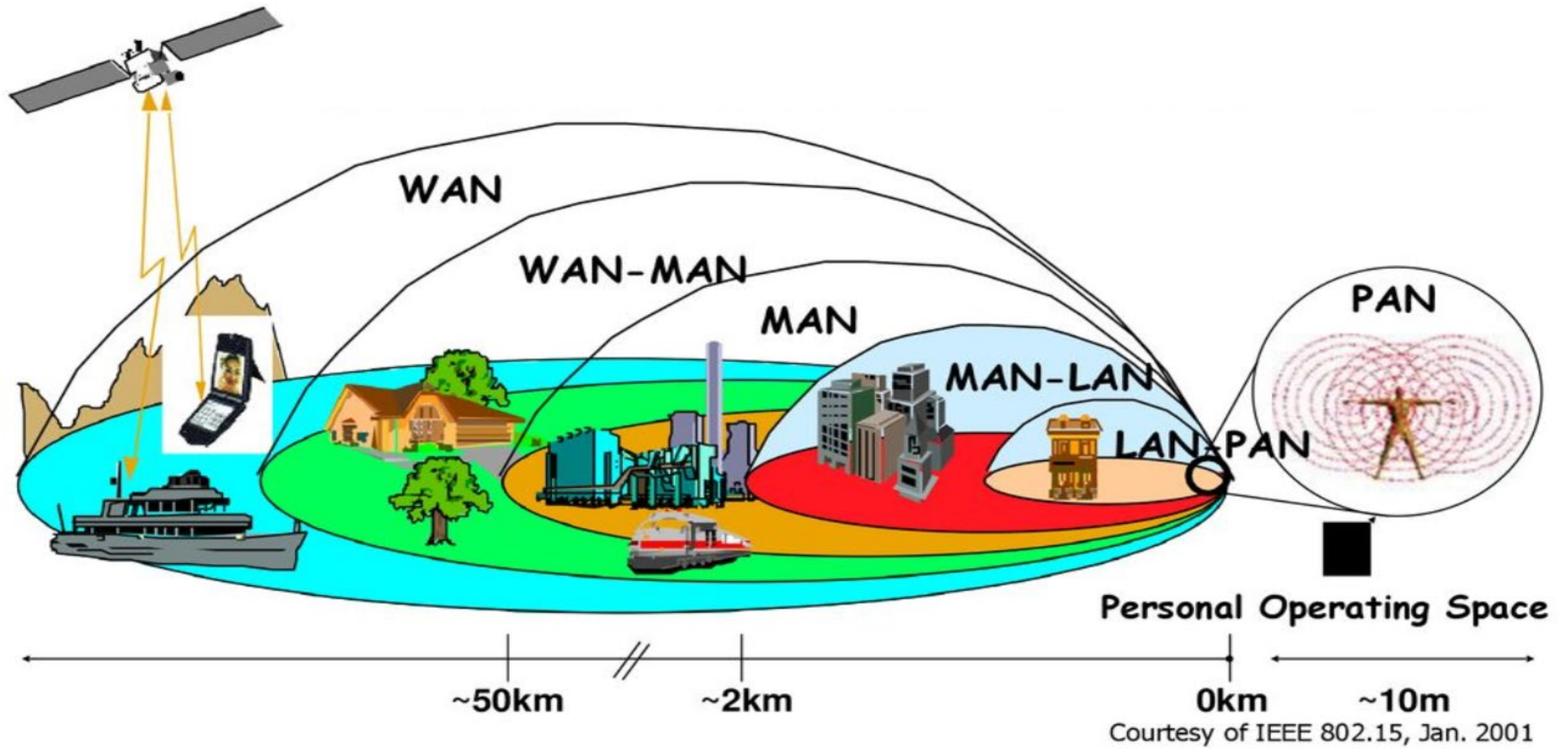
Metropolitan Area Network (MAN) covers a geographical area larger than LAN but smaller than WAN.

It can be considered as a network of LANs.



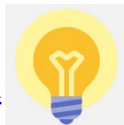
Wide Area Network (WAN)





Undersea Cables

<https://youtu.be/IIAJJI-qG2k>



Network Types

https://youtu.be/4_zSIXb7tLQ

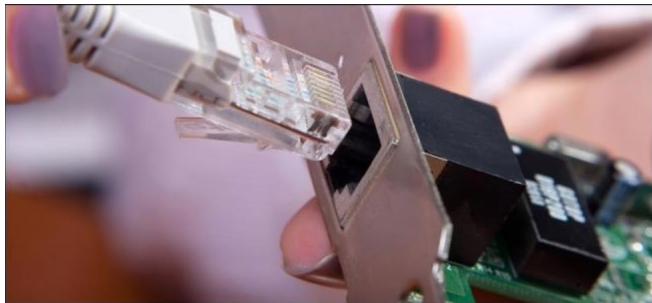


Part 1b : Network Infrastructure

Network Device 1 : A Switch

Network Switch and MAC Address

- A switch is a layer 2 network device that connects and allows a group of devices to communicate using their unique Media Access Control (MAC) addresses of their Network Interface Card (NIC).
- 48 bits hexadecimal = 2^{48} = approx. 2.8×10^{14} unique MAC addresses
- The switch maintains a table of the MAC addresses of the devices connected to the its ports.



MAC: 00-16-EA-06-6C-3E

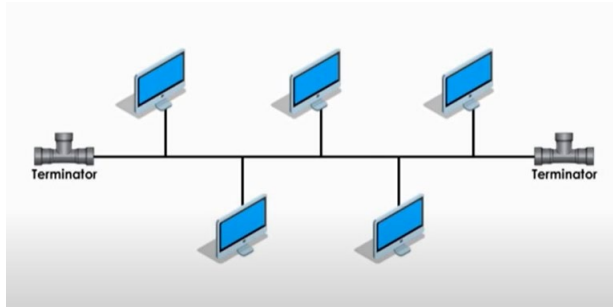
- 4/8/16/24/48 ports
- Ethernet/Fiber



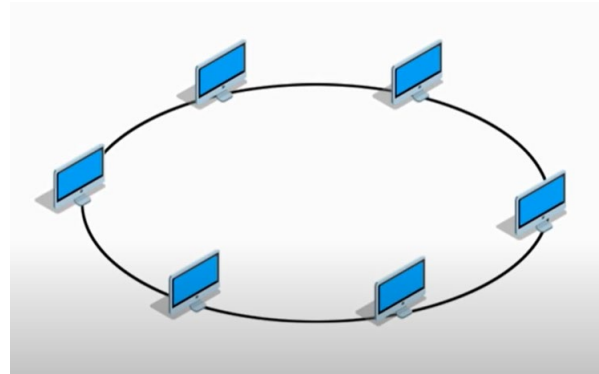
What is MAC address?

<https://youtu.be/UrG7RTWIJak>

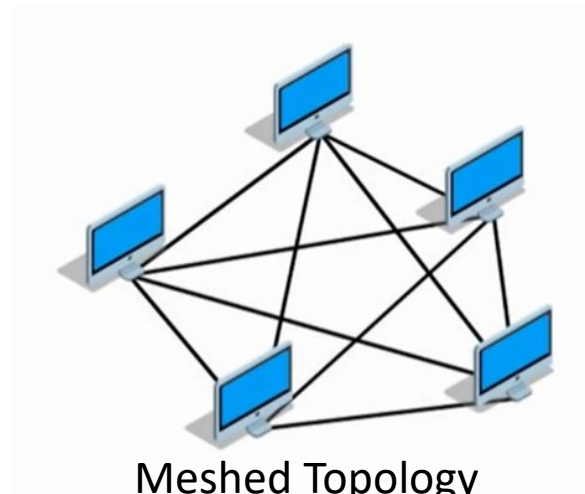
Topology: How to connect a group of devices?



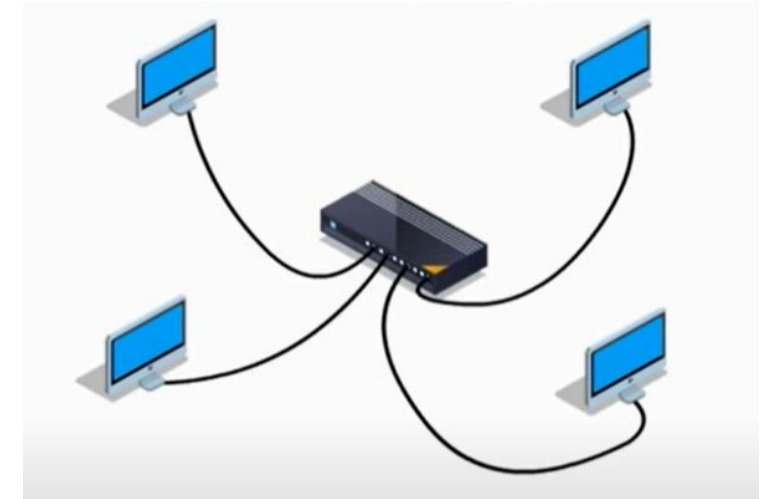
Bus Topology



Ring Topology



Meshed Topology



The **star topology** is most commonly used in a LAN where each node's NIC is connected directly to a network switch.



7 LAN Topologies

<https://youtu.be/T5xQJk-Aamk>

Hands-on 2:

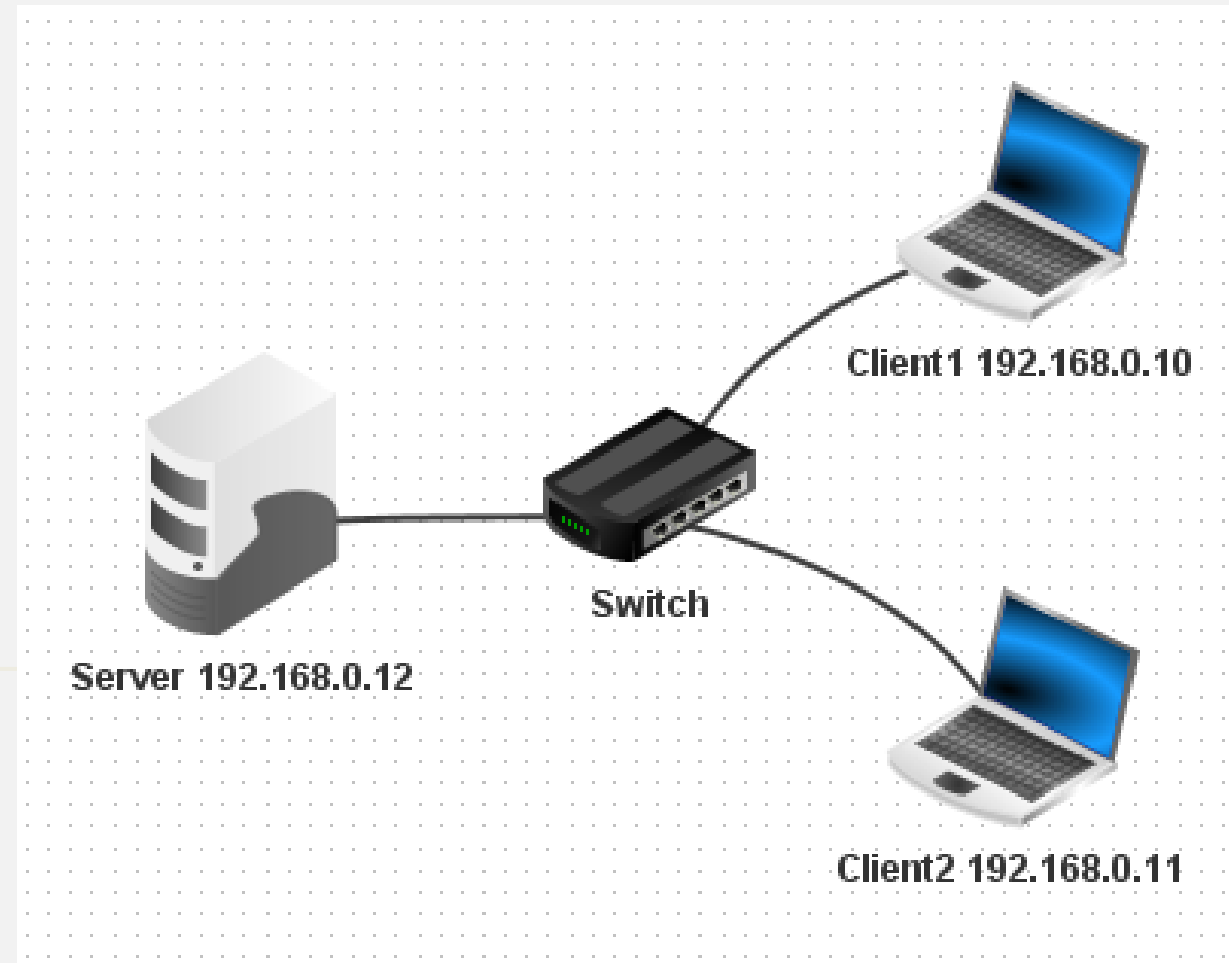
Construct a Local Area Network (LAN)

Construct a Server-Clients Network



Connect a Server to two client Notebooks using a network switch.

Configure the IP address of the Server as 192.168.0.12 and using the same subnet mask 255.255.255.0



Source Address Table (SAT) of the Network Switch



1. Click on the switch to check the SAT table.

MAC	Port
It is empty!	

2. Using the Command Line in Client1 and type Ping 192.168.0.11 to test the connection with Client2.

Click on the switch to check the SAT table.

These are the MAC addresses of Client1 and Client2.

MAC	Port
0F:7B:CC:D1:BB:11	Port 0
54:8C:B1:E6:C6:AC	Port 1

3. Test the connection with the Server by typing Ping 192.168.0.12

Click on the switch to check the SAT table.

MAC	Port
DF:35:23:B5:82:12	Port 2
0F:7B:CC:D1:BB:11	Port 0
54:8C:B1:E6:C6:AC	Port 1

This is the MAC address of the Server.



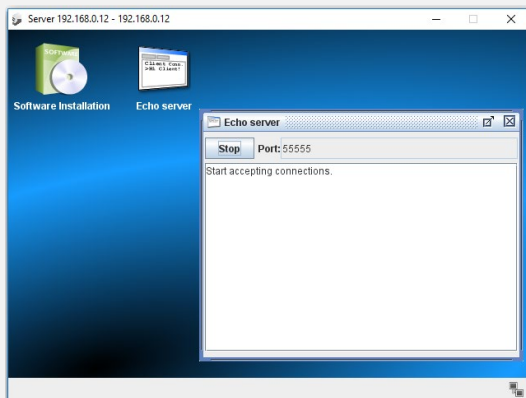
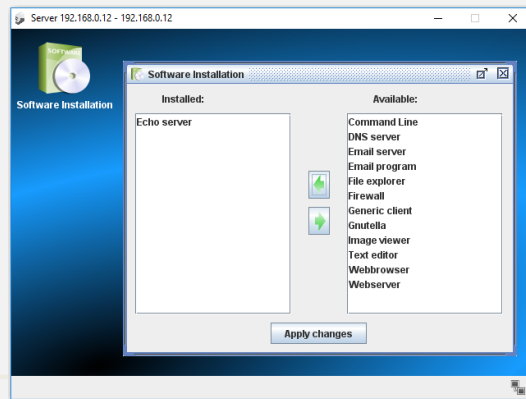
How a switch builds Source Address Table dynamically?

<https://youtu.be/DukuFSZH9Qw>

Install Echo Server and Generic Client for Messaging

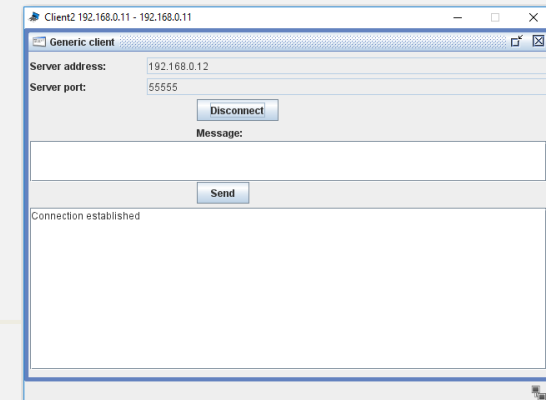
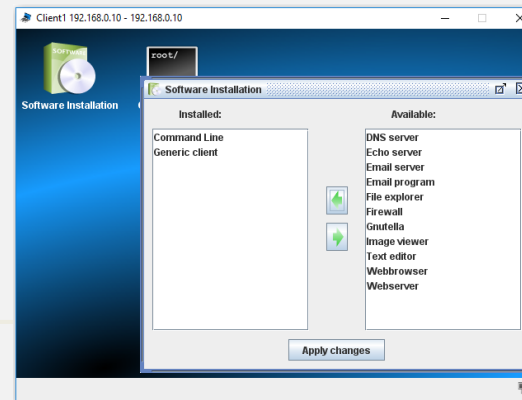


1. Click on the Server to install and Start the **Echo Server** using port 55555.



2. Install the **Generic Client** on both the client Notebooks.

Connect to the Server at 192.168.0.12 and port 55555.



3. Send any message to the Server and it will echo back. Observe the **green lines** simulating the transmission over the cables.

Server - Client Network

Advantages	Disadvantages
All files are stored in a central location	A specialist network operating system is needed
Network peripherals are controlled centrally	The server is expensive to purchase
Backups and network security is controlled centrally	Specialist staff such as a network manager is needed
Users can access shared data which is centrally controlled	If any part of the network fails a lot of disruption can occur

Source: https://www.teach-ict.com/gcse_new/networks/peer_peer/miniweb/pg3.htm

Part 1b : Network Infrastructure

Network Device 2 : A Router

Router

- A router is a layer 3 network device that routes the data from one network to another network or the Internet using the Internet Protocol (IP) addresses.
- A router is essentially the **gateway** of the network.
- The router maintains a routing table of the known **IP addresses** and the possible paths.

Enterprise Router



Hybrid Router (include Switch and Wifi)



4-port Switch (RJ45)



Hub, Switch, & Router Explained - What's the difference?

https://youtu.be/1z0ULvg_pW8

Hands-on 3:

Connecting Local Area Networks (LANs)

Connecting two LANs



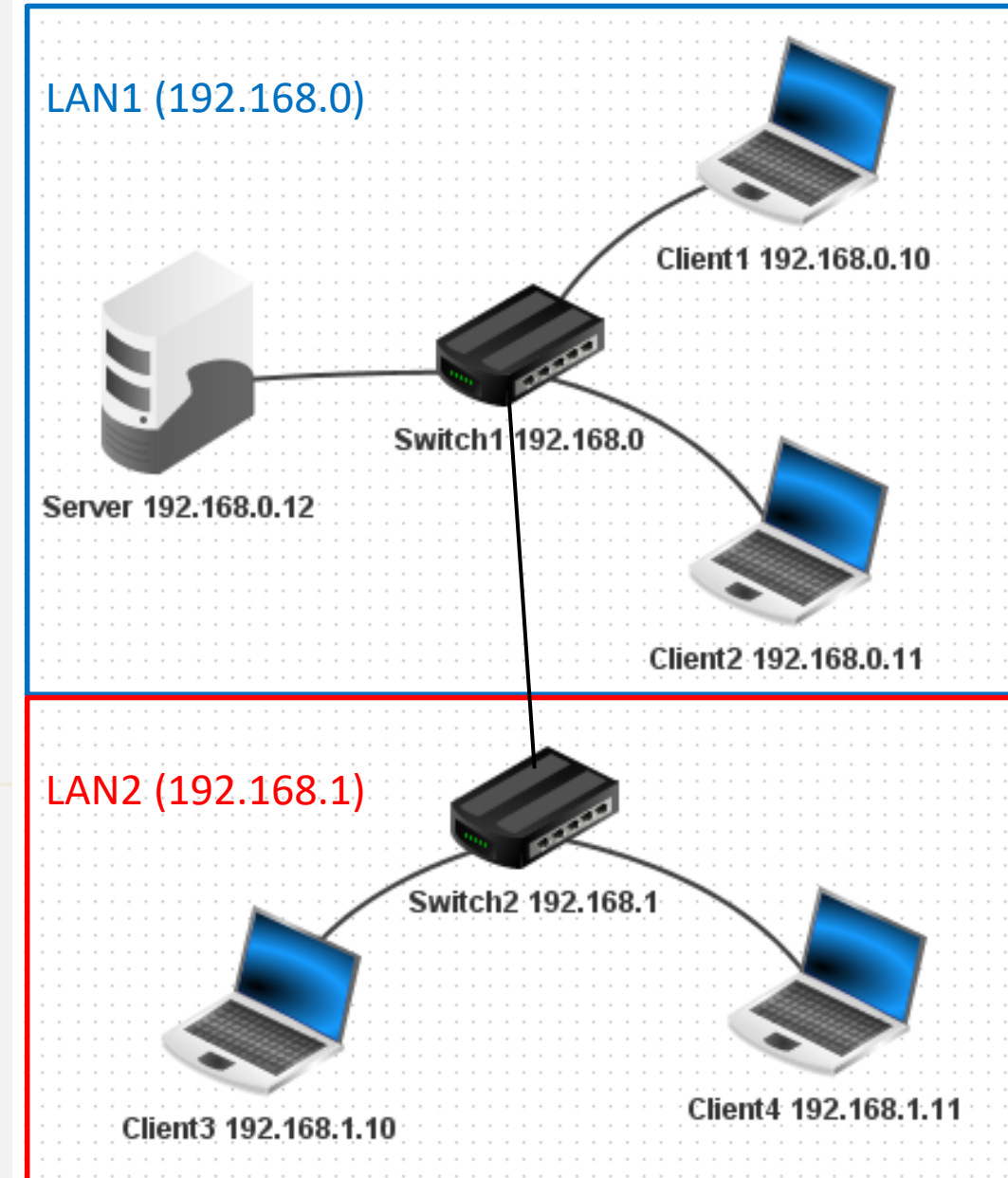
Construct a second network LAN2 which consists of a switch and two Notebooks; configure the Network ID of LAN2 as 192.168.1

Connect the two LANs using a cable to link the switches together.



Check for connectivity by *pinging* each other from the Command Line in the Notebooks.

There is no connection !!!



Connect LANs with a Router



Use a router to connect the two switches. Configure the IP addresses of the two Network Interface Cards (NICs) in the router :

NIC1 : 192.168.0.1

NIC2 : 192.168.1.1

Configure the **Gateway** for all the Notebooks and Server in the LANs:

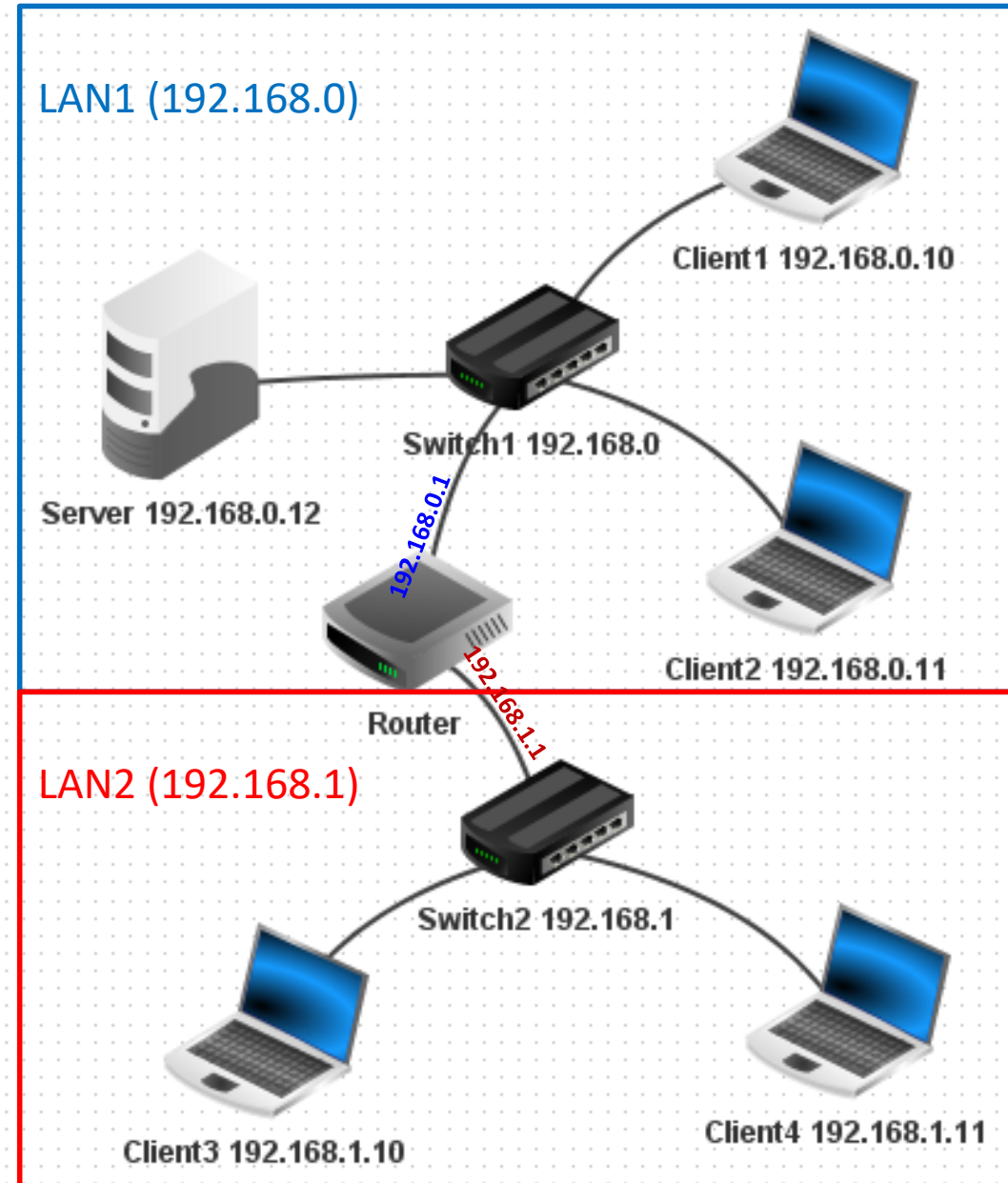
Gateway for LAN1 : 192.168.0.1

Gateway for LAN2 : 192.168.1.1



Ping for connectivity. Observe the **green lines** simulating the transmission over the cables.

There is connection !!!



Part 2 : Network Addressing

- MAC and IP Addressing

Sunny Classroom

In the TCP/IP world,
why do we use
both IP and MAC address?



Why both IP and MAC?

<https://youtu.be/oGoWgdlaOMI>



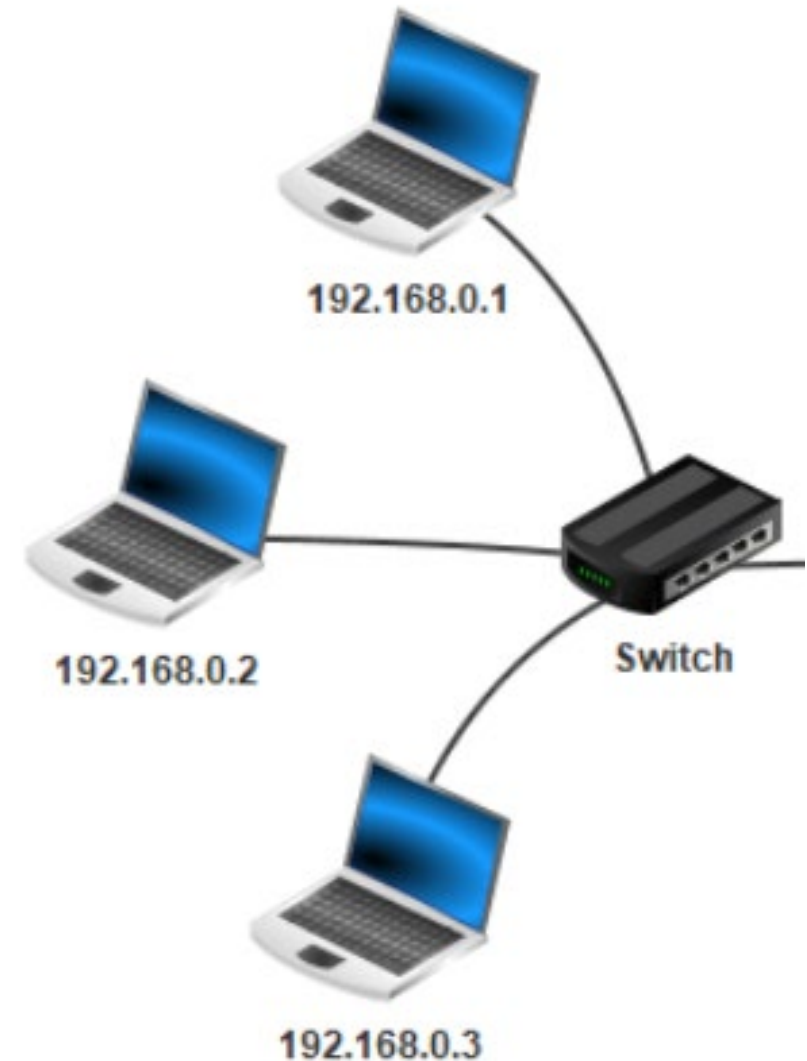
0:07 / 3:38

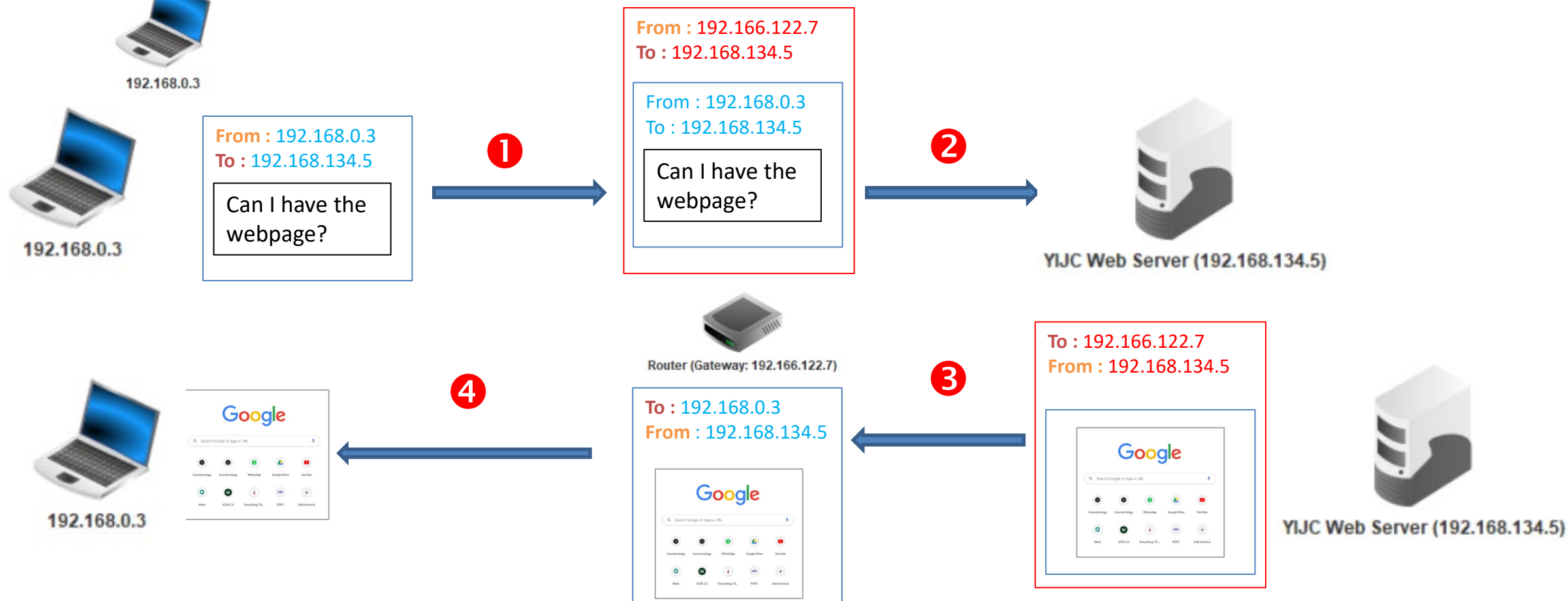
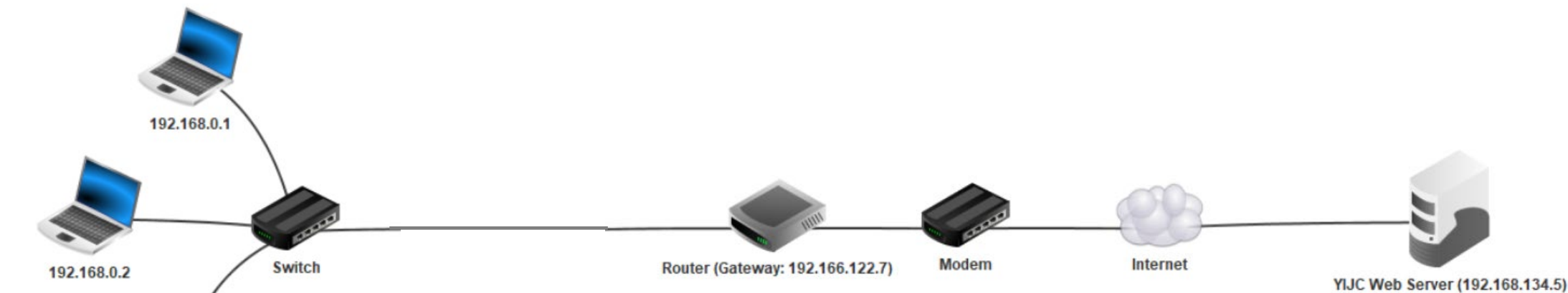


More about Private and Public IP Address

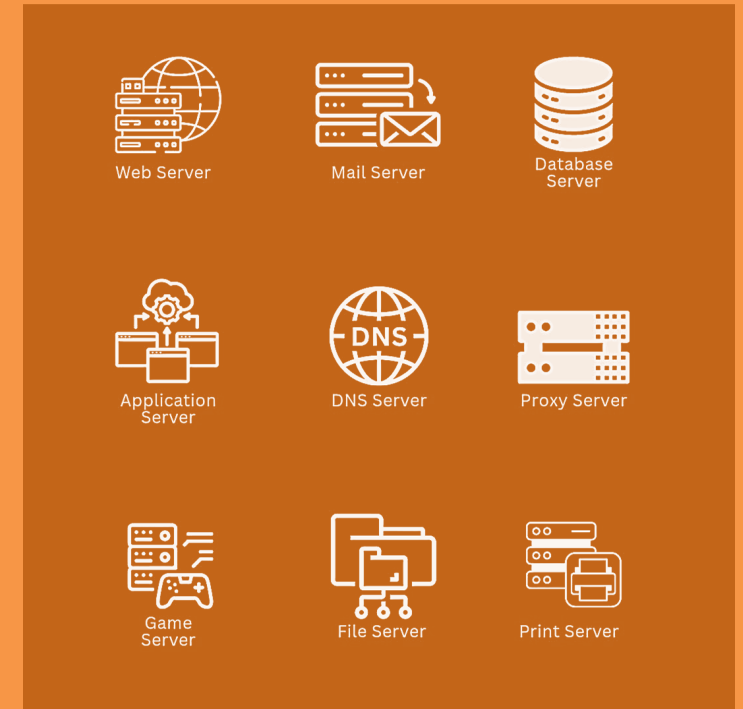
Local Area Network (LAN)

- Both Private and Public IP addresses look the same. For example 192.168.0.1, this is a 32-bit IP address, 4 sets of 8-bit number.
- The devices within a LAN are identified by their Private IP addresses.
- How Address Resolution Protocol (ARP) is used to find the MAC address of a device in a LAN from the IP address (<https://youtu.be/-bXsCweU1DA>)





Part 3a : About Servers

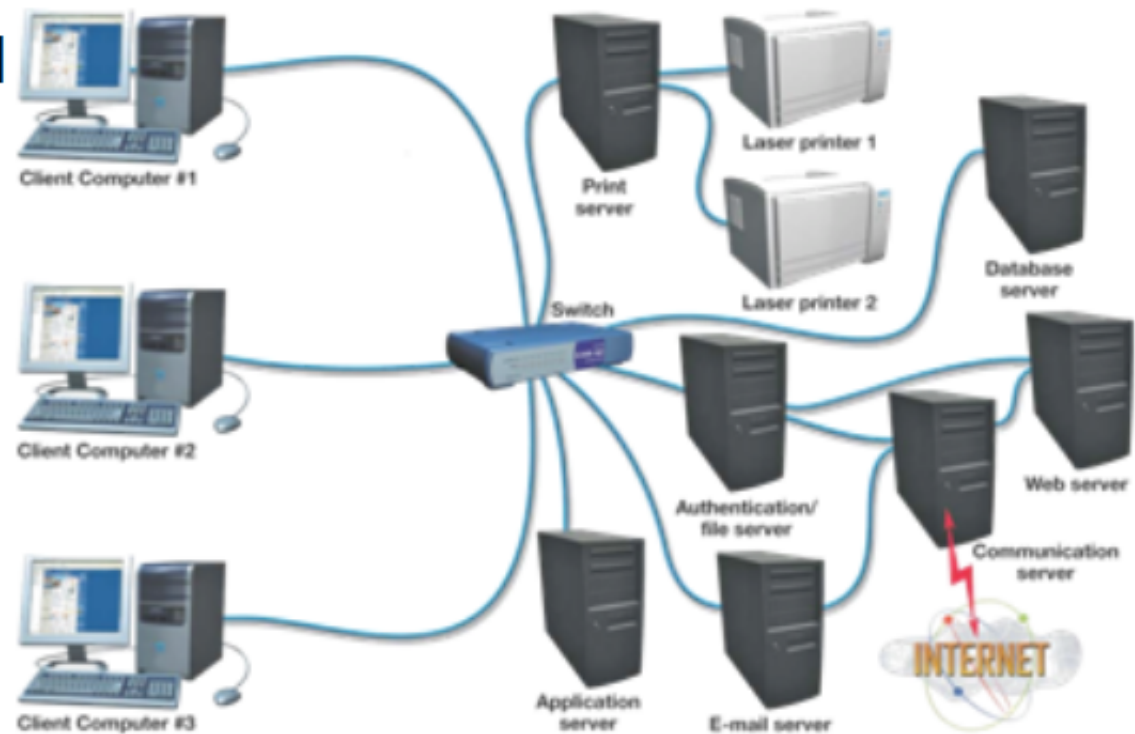


What is a Server?

<https://youtu.be/UjCDWCeHCzY>

Servers

- Number and type of servers depend on network size and workload
- Dedicated server
 - Performs one specific function
- Authentication server
 - Keeps track of network logins and services available
- File server
 - Stores and manages files



Dedicated Servers

- Communications server
 - Handles communications between networks including the Internet
 - Often the only device on the network directly connected to the Internet
- Web server
 - Hosts a Web site available through the Internet

Dedicated Servers

- Print server
 - Manages client-requested printing jobs
 - Creates print queue (prioritizes print jobs)
- Applications server
 - Acts as a storage area for application software
- Database server
 - Provides clients with access to database information
- E-mail server
 - Processes and delivers incoming and outgoing e-mail

Hands-on 4:

a. Installing Web Server



How does the internet works?

https://youtu.be/7_LPdttKXPc



The difference between the Internet and the Web

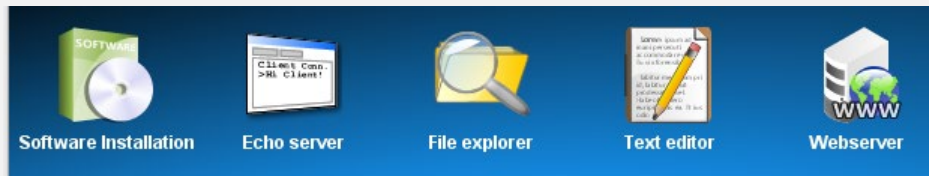
<https://youtu.be/UVkT59PmRdo>

Installing a Web Server

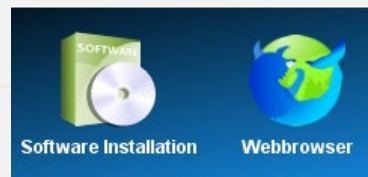
Use the network set up in **Hands-on 3** and configure the server to be a WWW Server.



Install the **Webserver**, File explorer and the Text editor on the WWW Server.

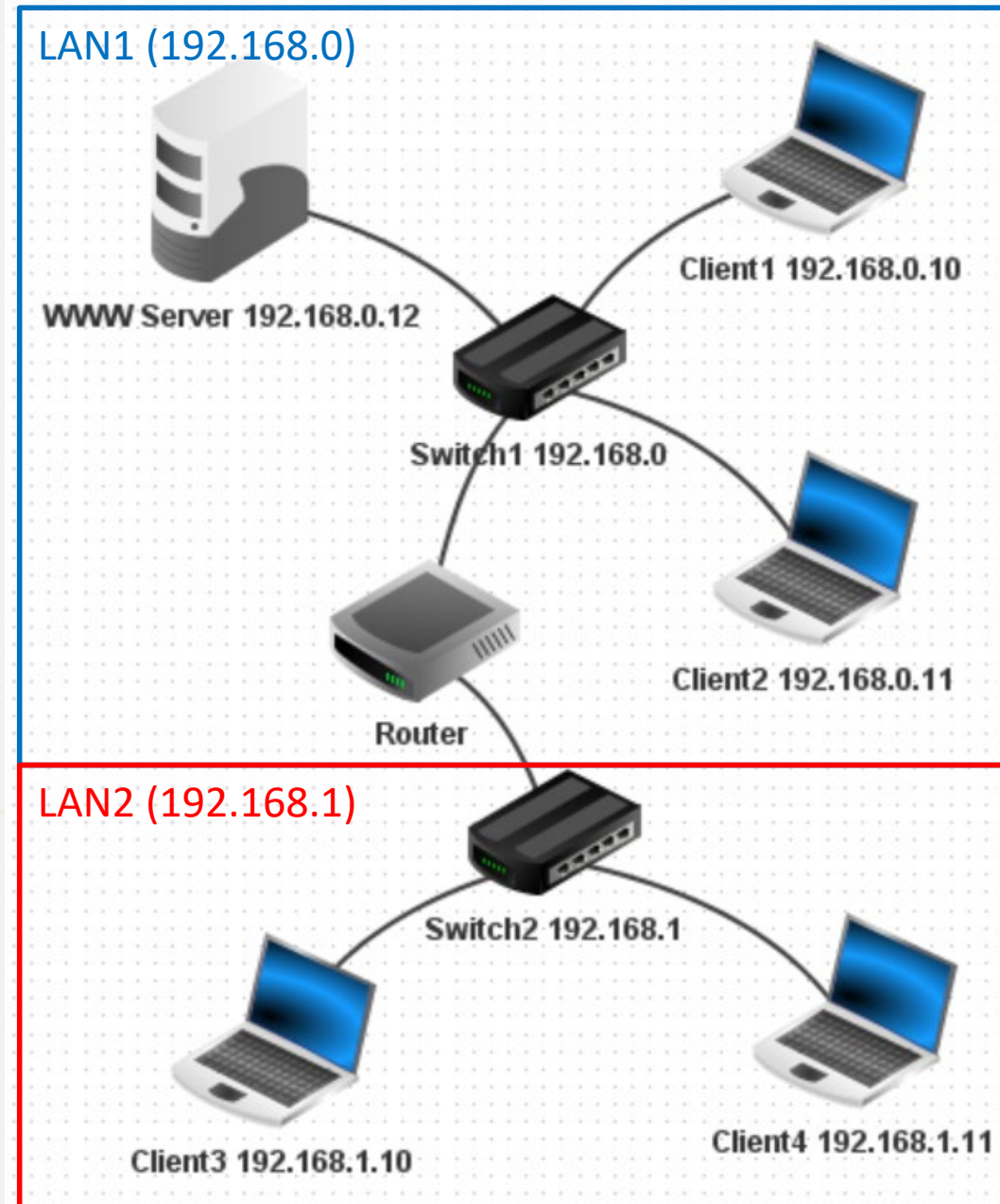


Install the **Webbrowser** on at least one of the Notebooks.

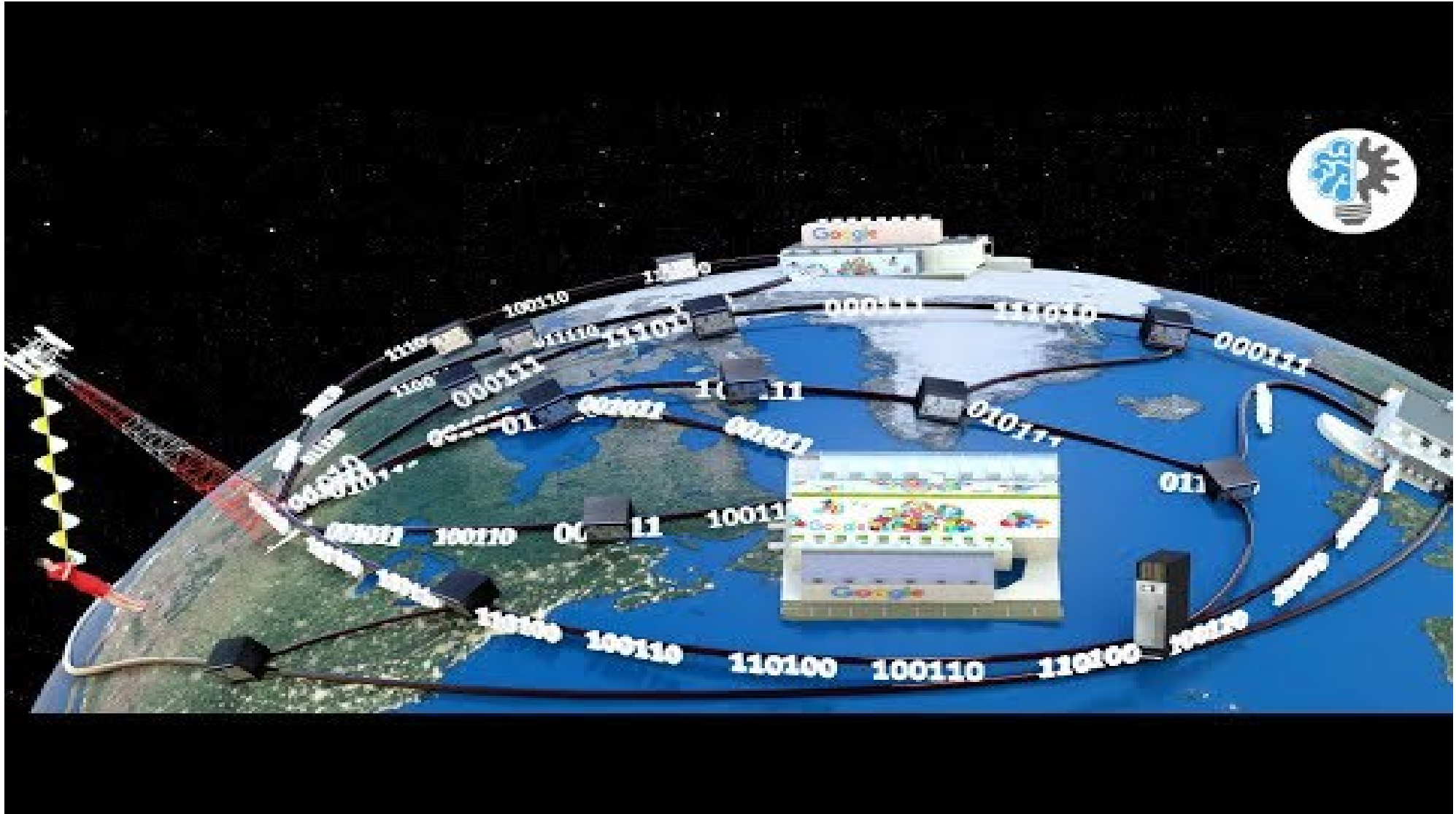


 the **Webserver** and access the website from the Notebook's **Webbrowser** using the url:

`http://192.168.0.12`



How does the Internet works?



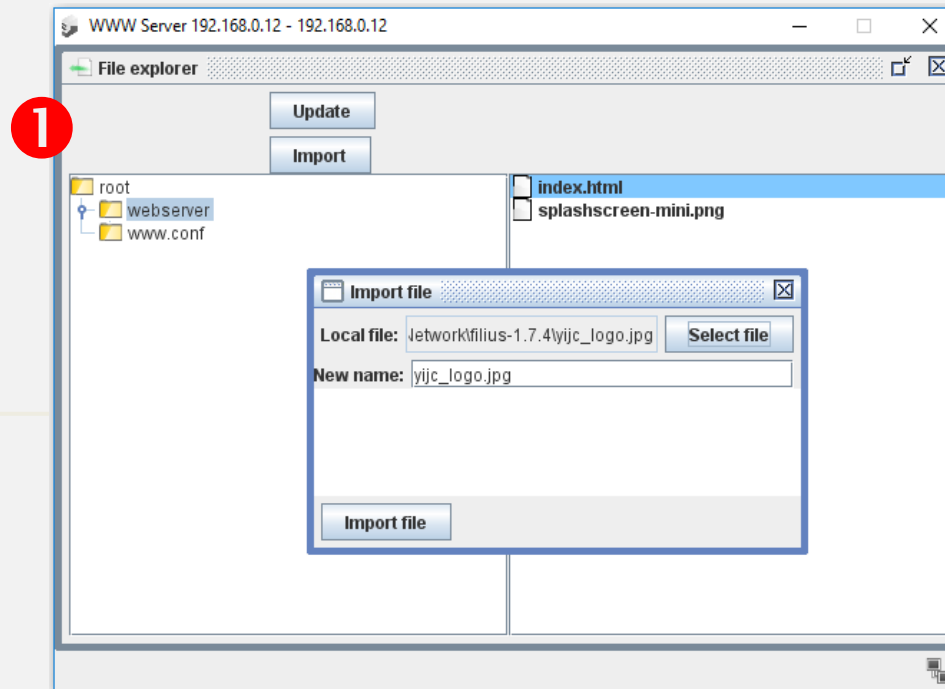
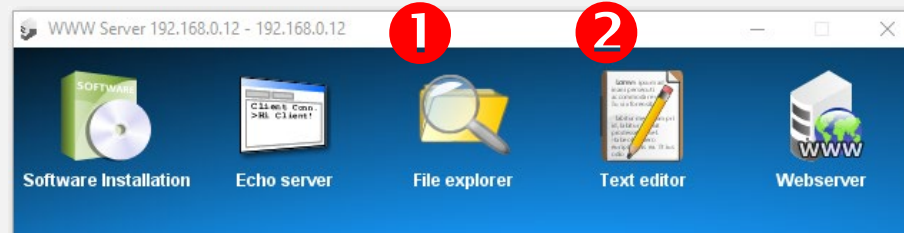
How does the Internet works? <https://youtu.be/x3c1ih2NJEg>

Try-It : Customise our Homepage

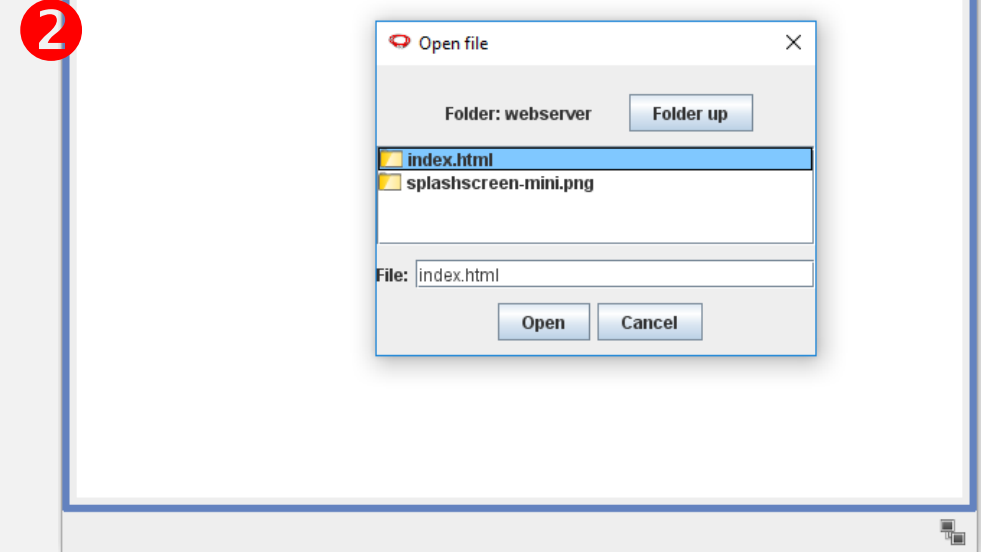


Using the Web
Server in
Hands-on 4

1. Upload `yijc_logo.jpg` using the **File explorer**.
2. Edit `index.html` using the **Text editor**.



Import the `yijc_logo.jpg` file into the **webserver** folder.



Using the Text editor to open the `index.html` file within the **webserver** folder.

Modify according to the next slide.

Try-It : Simple HTML Editing

The diagram illustrates the process of editing HTML files on a web server and viewing the results in a browser. It consists of four main components: two code editors for 'index.html' on a 'WWW Server 192.168.0.12' and two browser windows for 'Client4 192.168.1.11 - 192.168.1.11'.

Left Code Editor (index.html):

```
<html>
<head>
<title>Standardseite</title>
</head>
<body bgcolor="#ccddff" style="font-family:Verdana; text-align:center;">
<h2> FILIUS - Webserver </h2>
<p>Welcome at FILIUS!</p>
<p> This page has been delivered as default page.
Feel free to add your own contents! </p>
<p align="center">  </p>
<p> http://www.lernsoftware-filius.de </p>
</body>
</html>
```

Right Code Editor (index.html):

```
<html>
<head>
<title>HomePage</title>
</head>
<body bgcolor="#ccddff" style="font-family:Verdana; text-align:center;">
<h2> Yishun Innova Junior College </h2>
<p>Welcome to YIJC !</p>
<p> Congratulation! Your Web Server is up and running ... </p>
<p align="center">  </p>
Click <a href="http://yijc.moe.edu.sg"> HERE </a> to visit the homepage.
</body>
</html>
```

Left Browser Window (Standardseite):

Url for the Webserver: 192.168.0.12

Standardseite

http://www.yijc.edu.sg/

FILIUS - Webserver

Welcome at FILIUS!

This page has been delivered as default page. Feel free to add your own contents!

Internetworking

http://www.lernsoftware-filius.de

Right Browser Window (Standardseite):

Standardseite

http://www.yijc.edu.sg/

Yishun Innova Junior College

Welcome to YIJC !

Congratulation! Your Web Server is up and running ...



YISHUN INNOVA
JUNIOR COLLEGE

Click [HERE](#) to visit the homepage.

Part 3b : About Servers

Hands-on 5:

b. Installing a Domain Name Server (DNS)

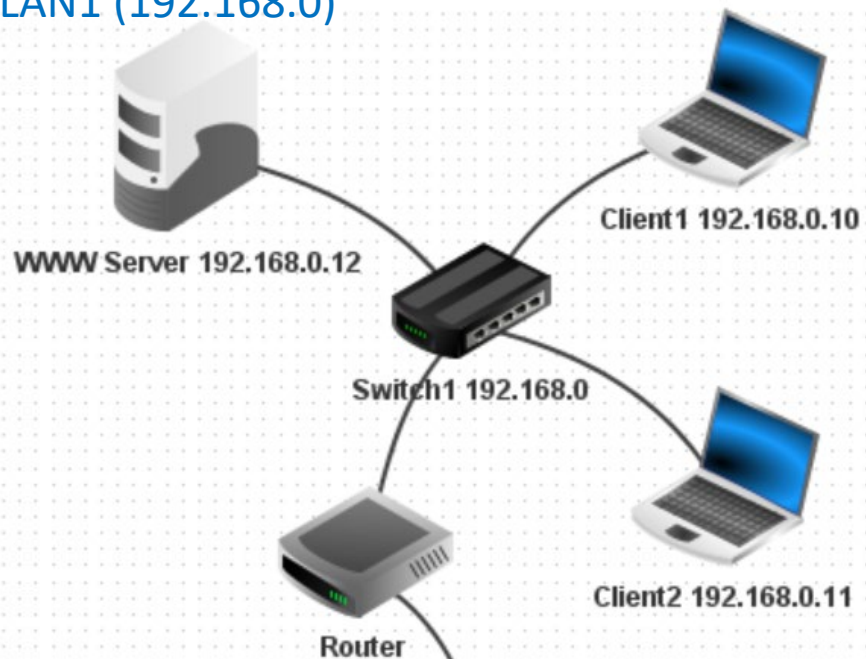


How the Domain Name Server (DNS) works?

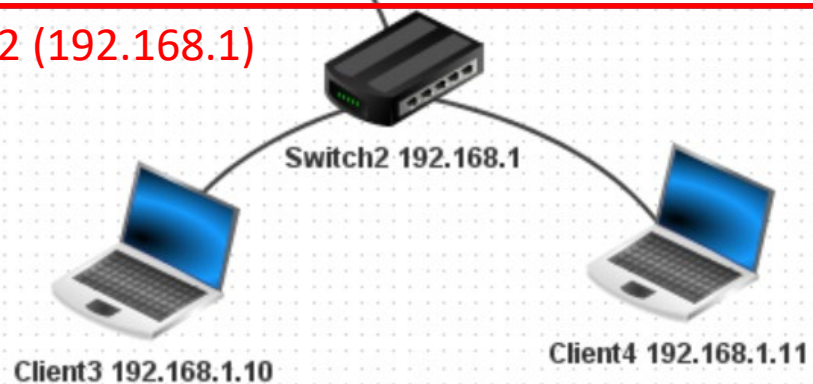
<https://youtu.be/mpQZVYPuDGU>

Installing a DNS Server

LAN1 (192.168.0)



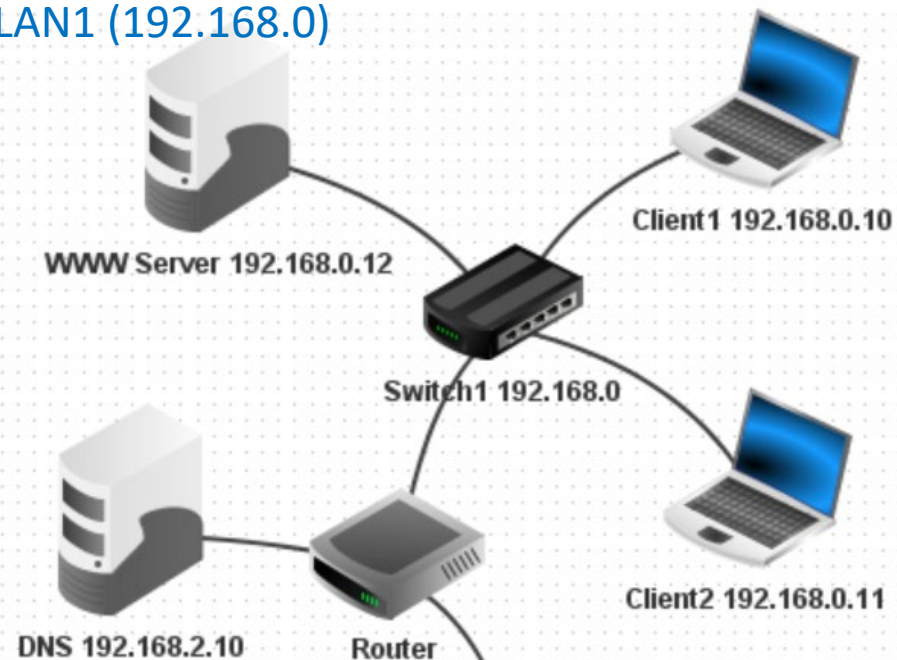
LAN2 (192.168.1)



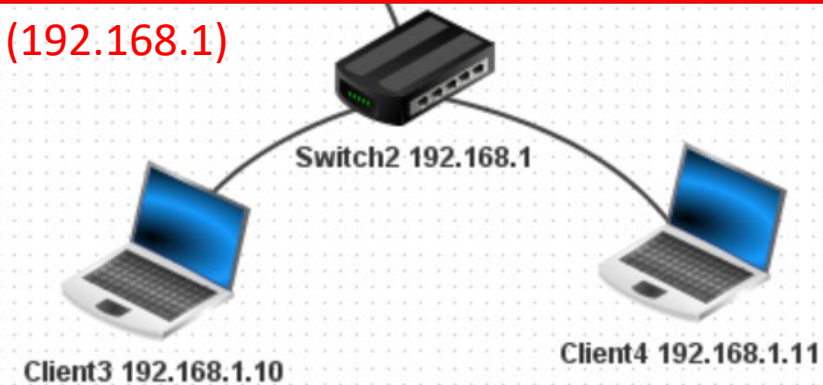
Modify
Hands-on 4
Network

1. Add a NIC in the Router.
2. Add Server.
3. Setup the DNS Server.
4. Configure the DNS address for all other Server and Notebooks'.
5. Test the DNS Server.

LAN1 (192.168.0)



LAN2 (192.168.1)



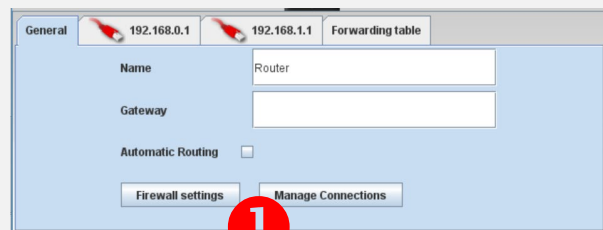
1. Add a NIC in the Router

Click on the Router to add an additional NIC.

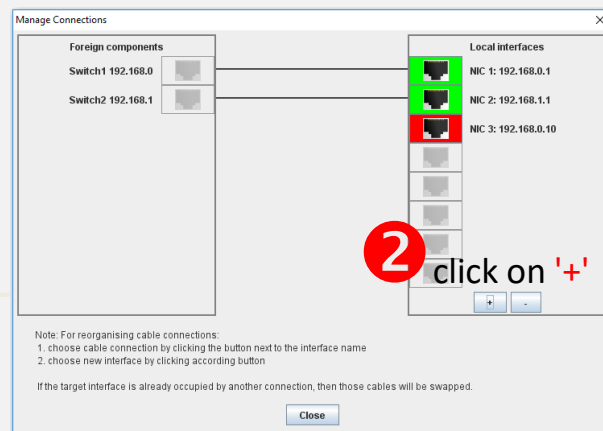


Modify
Hands-on 4
Network

1. Add a NIC in the Router.
2. Add Server.
3. Setup the DNS Server.
4. Configure the DNS address for all other Server and Notebooks'.
5. Test the DNS Server.



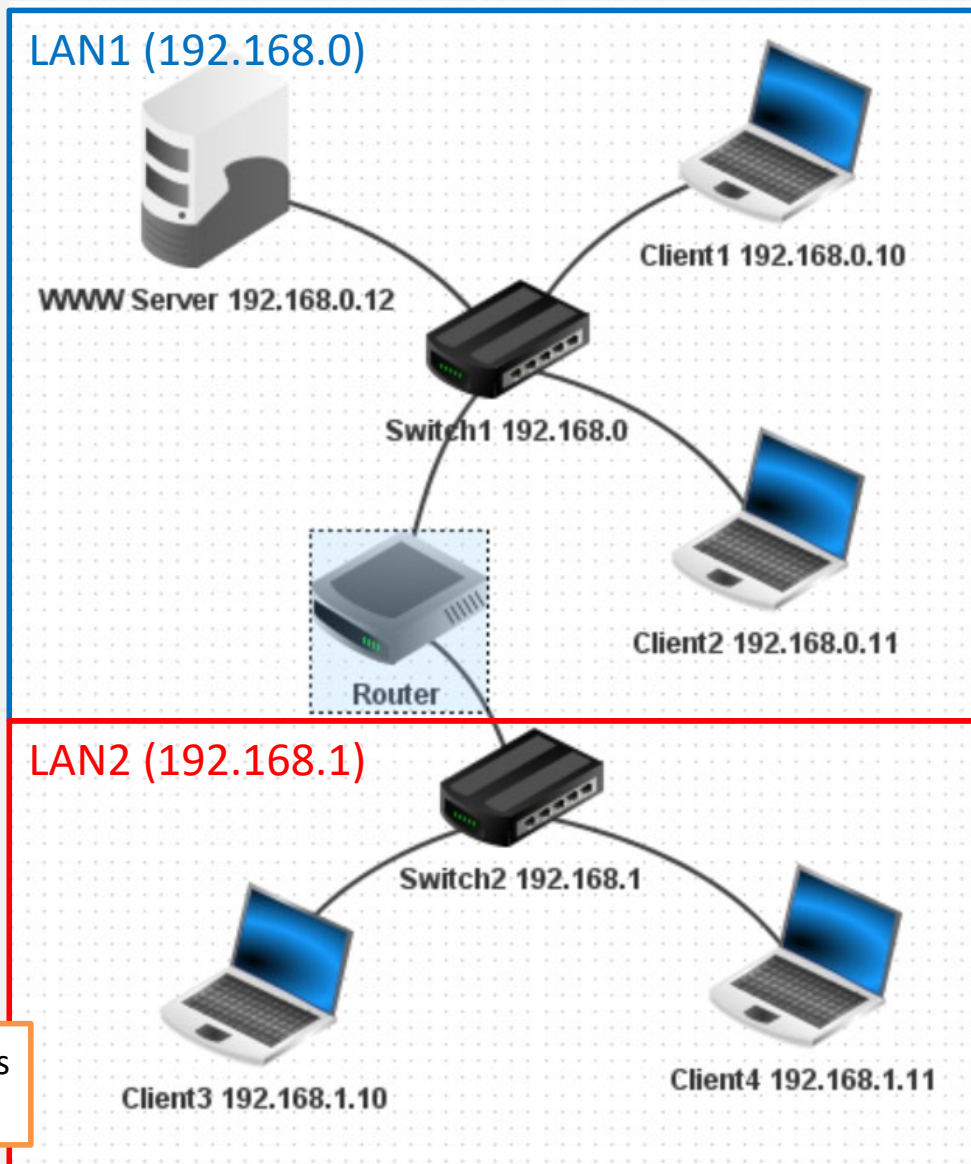
click on 'Manage Connections'



click on '+'



Configure NIC IP address
as: 192.168.2.1



2. Add a Server

- 1 Drag-and-drop a Server on the Workspace. Configure the IP address of this Server as:

192.168.2.10

Name	DNS 192.168.2.10	☐ Use IP address as Name
MAC Address	16:D6:AC:40:6D:B1	☐ Use DHCP for configuration
IP address	192.168.2.10	DHCP server setup
Netmask	255.255.255.0	
Gateway		
Domain Name Server		

Configure the IP address as: 192.168.2.10

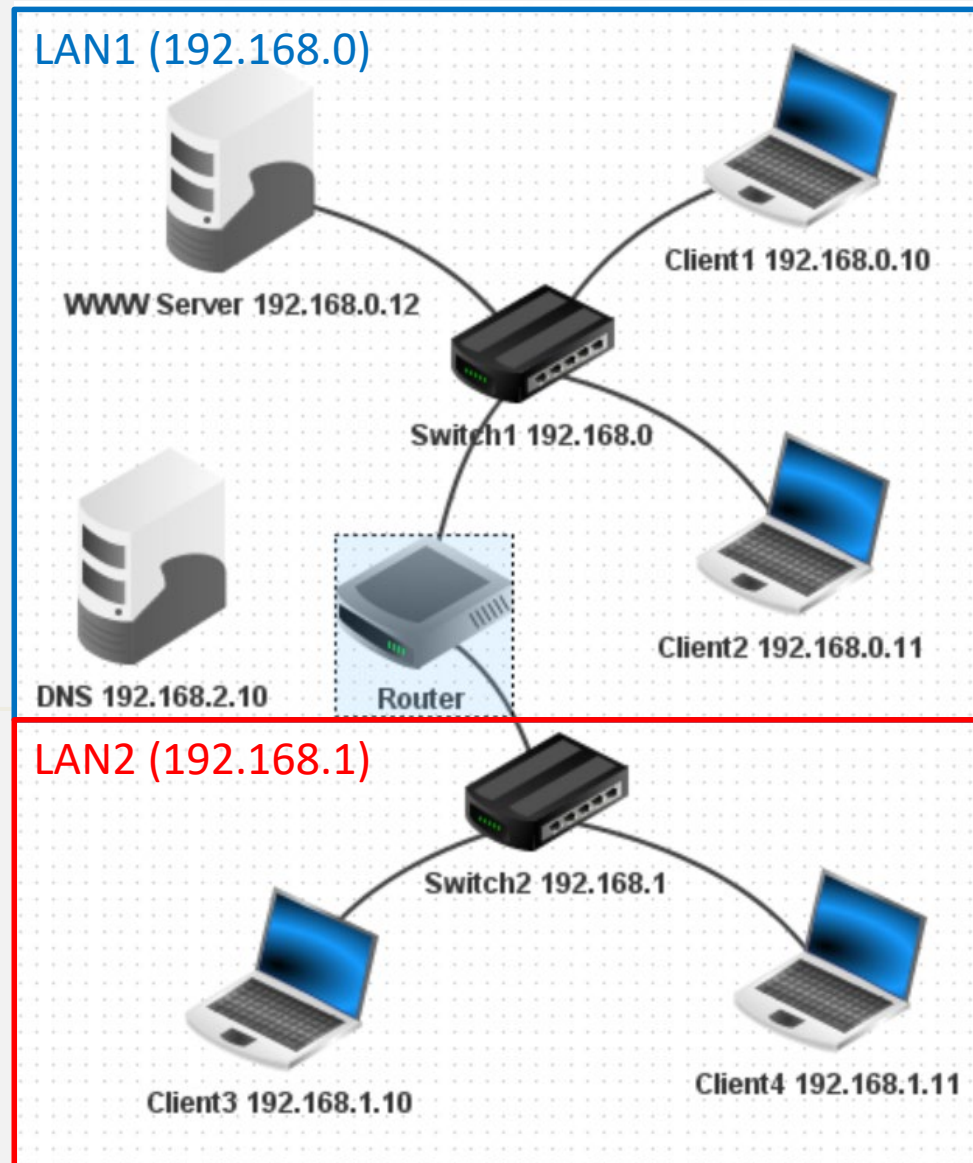
3 Gateway :
192.168.2.1

4



Connect the Server to the Router with patch cable.

1. Add a NIC in the Router.
2. Add Server.
3. Setup the DNS Server.
4. Configure the DNS address for all other Server and Notebooks'.
5. Test the DNS Server.



3. Setup DNS Server

1. Add a NIC in the Router.
2. Add Server.
3. Setup the DNS Server.
4. Configure the DNS address for all other Server and Notebooks'.
5. Test the DNS Server.



1 In simulation mode, click on the Server to install the DNS Server program.



2 Launch the DNS server program to add a new entry for the WWW Server:

3 Domain name: **www.yjc.edu.sg**
IP address: **192.168.0.12**

4

DNS 192.168.2.10 - 192.168.2.10

DNS server

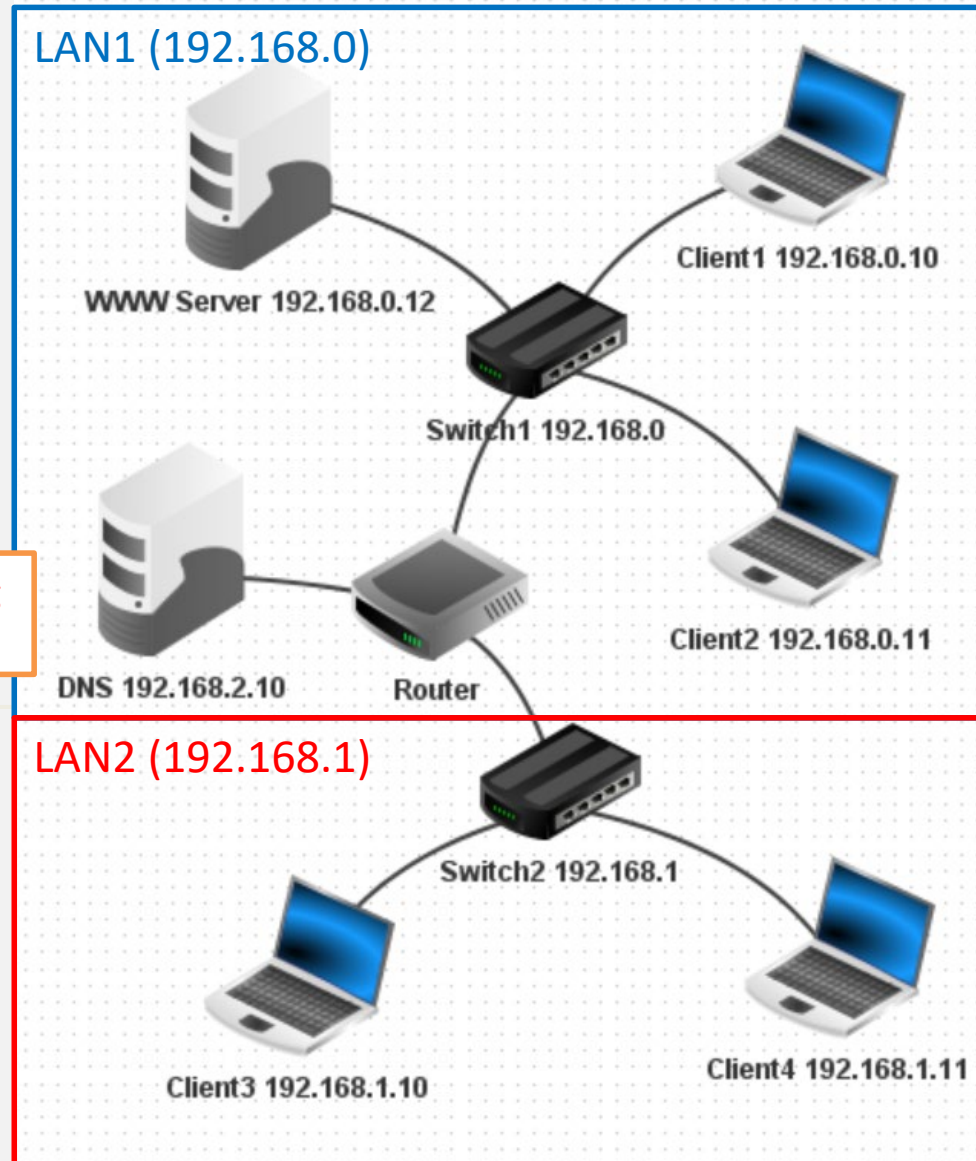
☐ Enable recursive domain resolution

Domain name:

IP address:

Domain name	IP address

Click to start the DNS Server.



4. Configure all the Notebooks and Server

1. Add a NIC in the Router.

2. Add Server.

3. Setup the DNS Server.

4. Configure the DNS address for all other Server and Notebooks'.

5. Test the DNS Server.



1 In development mode, click on every Notebooks and other Server to configure the Domain Name Server's IP address as:

192.168.2.10

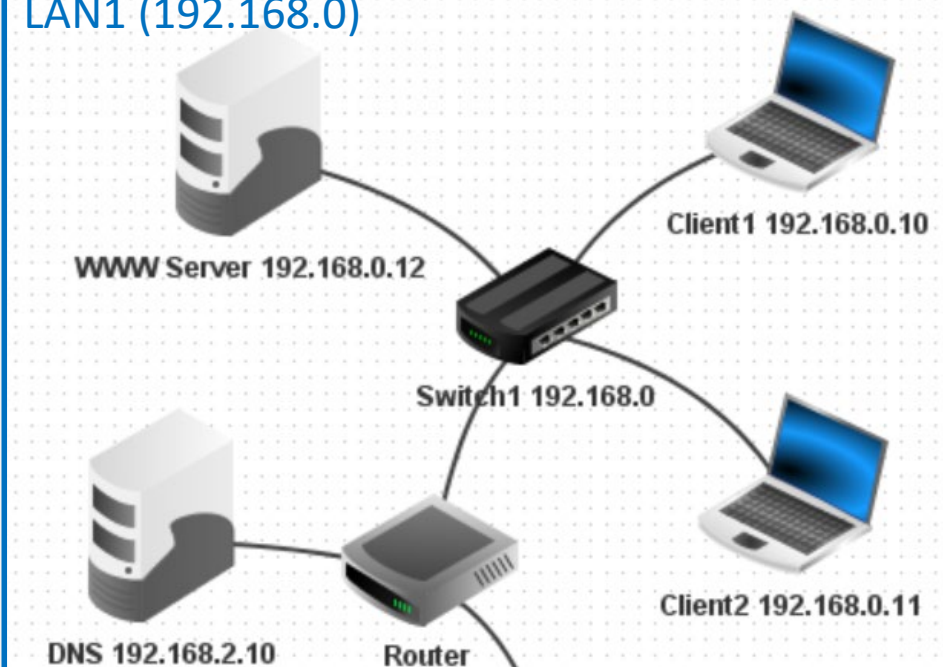
Name	Client1 192.168.0.10	☐ Use IP address as Name
MAC Address	0F:7B:CC:D1:BB:11	☐ Use DHCP for configuration
IP address	192.168.0.10	DHCP server setup
Netmask	255.255.255.0	
Gateway	192.168.0.1	
Domain Name Server	192.168.2.10	

DNS Server IP address:
192.168.2.10

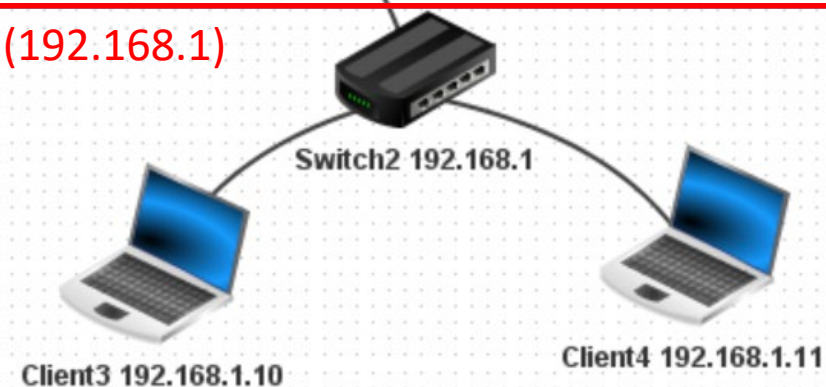
2

3 The DNS Server for the network is now ready for testing.

LAN1 (192.168.0)



LAN2 (192.168.1)

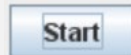


5. Test the network with the DNS Server

1. Add a NIC in the Router.
2. Add Server.
3. Setup the DNS Server.
4. Configure the DNS address for all other Server and Notebooks'.
5. Test the DNS Server.

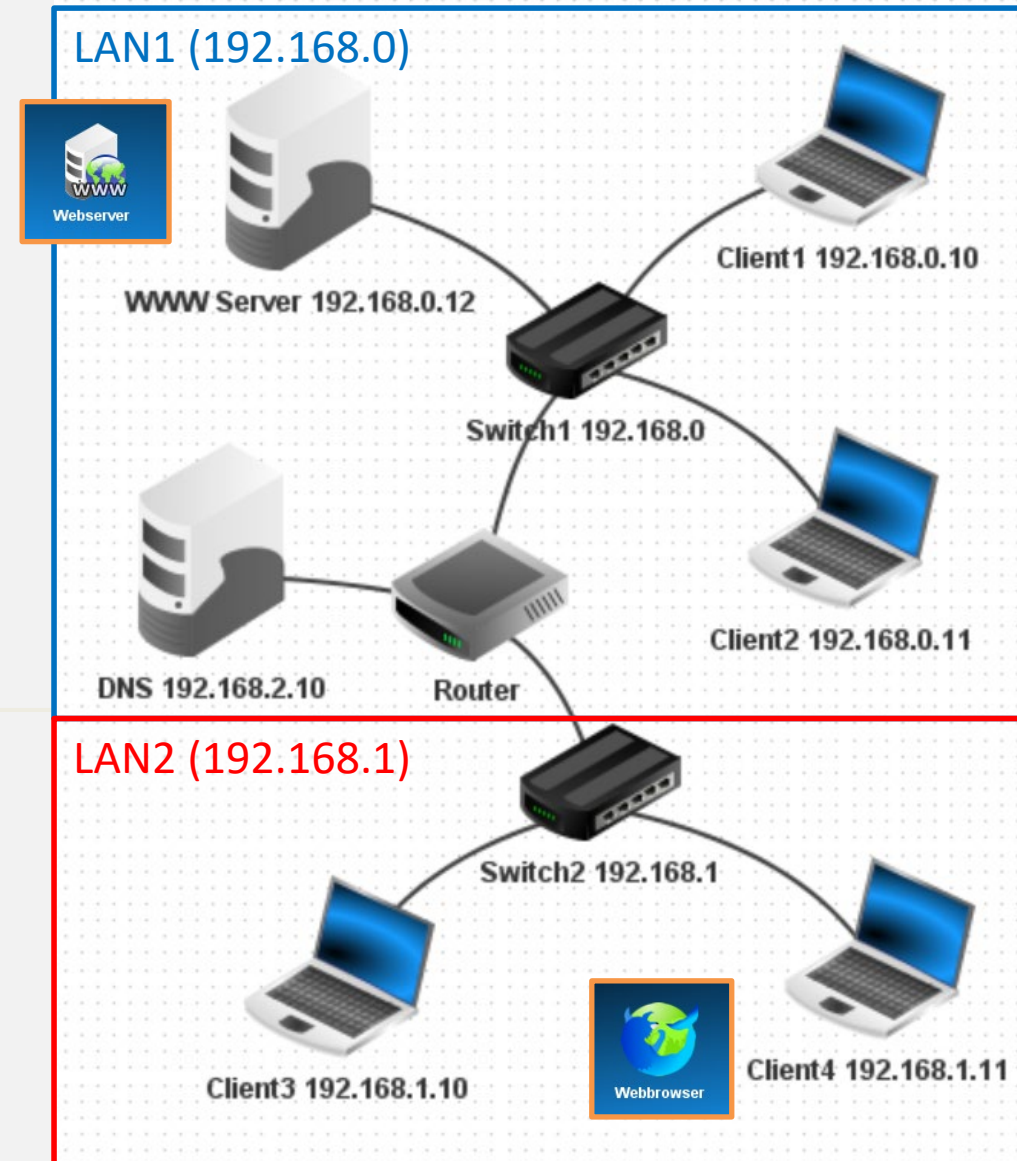


In Hands-on 4, the Webserver and Webbrowser have been installed and tested.



the **Webserver** and access the website from the Notebook's **Webbrowser** using the following url:

`http://www.yijc.edu.sg`



Hands-on 6:

c. Installing DHCP Server



DHCP Explained – Dynamic Host Configuration Protocol

<https://youtu.be/e6-TaH5bkjo>

Using a DHCP Server



Set up the LANs as shown. Configure the two DHCP Servers as follows:

Name	DHCP Server 192.168.0.10	☐ Use IP address as Name
MAC Address	DF:35:23:B5:82:12	☐ Use DHCP for configuration
IP address	192.168.0.10	DHCP server setup
Netmask	255.255.255.0	
Gateway	192.168.0.1	
Domain Name Server		

Name	DHCP Server 192.168.1.10	☐ Use IP address as Name
MAC Address	EF:26:B9:A2:98:4E	☐ Use DHCP for configuration
IP address	192.168.1.10	DHCP server setup
Netmask	255.255.255.0	
Gateway	192.168.1.1	
Domain Name Server		

Gateway address is the Router's NIC IP address

DHCP server setup

Base Settings Static Address Assignment

Lower bound of address: 192.168.0.11

Upper bound of address: 192.168.0.99

Netmask: 255.255.255.0

Gateway: 192.168.0.1

DNS server: 0.0.0.0

☒ Manual configuration

☒ Activate DHCP

OK



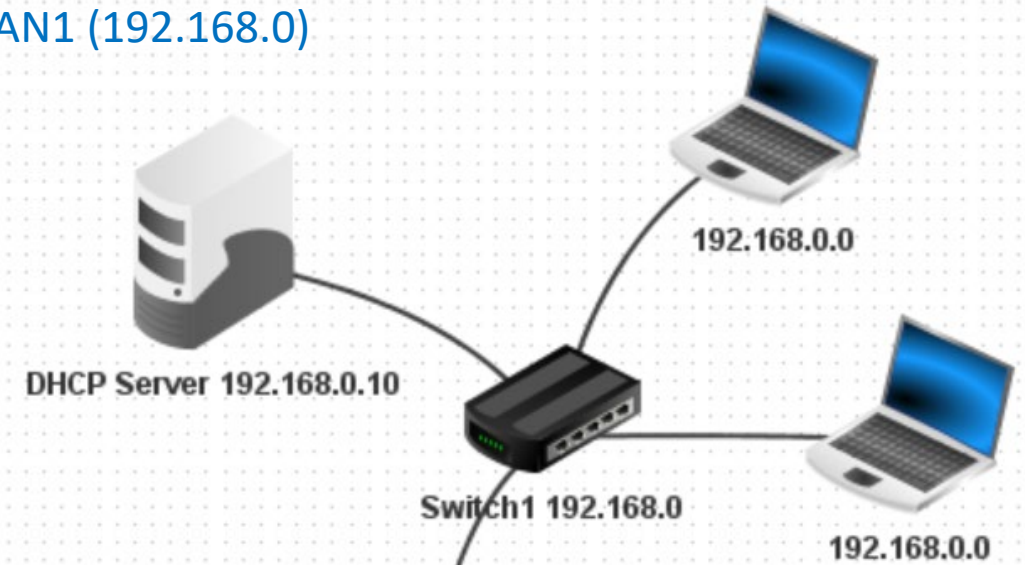
Configure all the Notebooks' IP addresses as 192.168.0.0

☒ Use IP address as Name ☒ Use DHCP for configuration

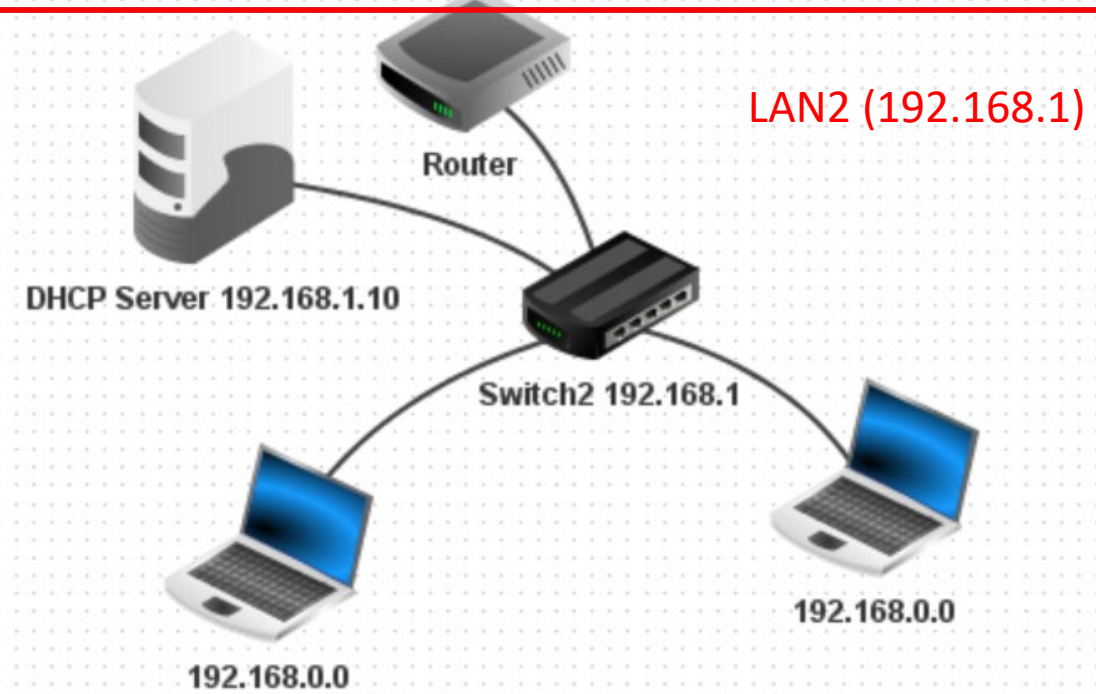


Observe the simulating **green lines** when the IP addresses are auto-generated.

LAN1 (192.168.0)



LAN2 (192.168.1)



Hands-on 7:

d. Installing a Mail Server



What is SMTP - Simple Mail Transfer Protocol

<https://www.youtube.com/watch?v=PJo5yOtu7o8>



POP3 vs IMAP – What's the difference

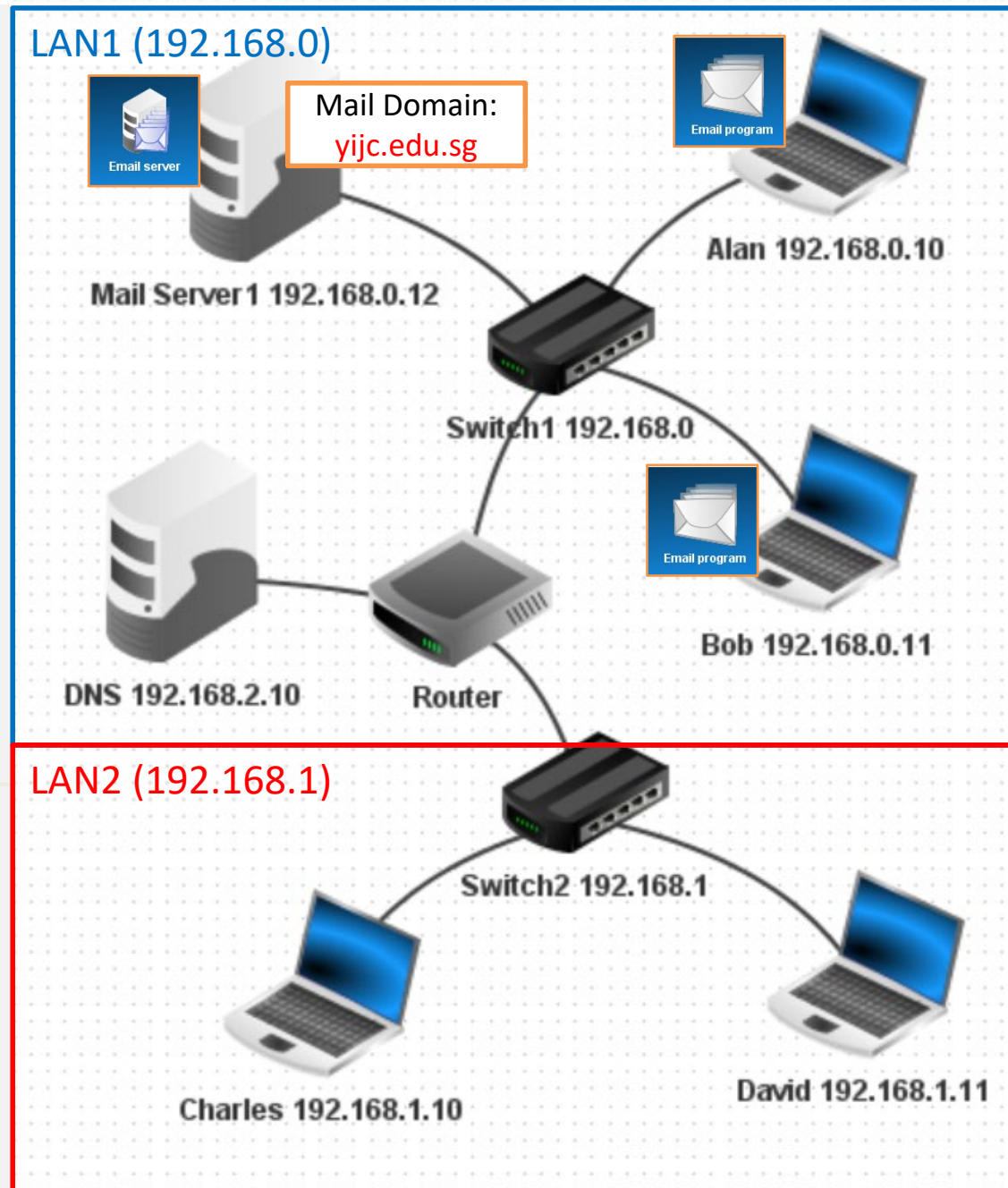
<https://www.youtube.com/watch?v=SBaARws0hy4>



How does a Mail Server works?

If Alan (alan@yijc.edu.sg) sends an email to Bob (bob@yijc.edu.sg) :

1. The email goes from Alan's laptop to the switch1 and then to the YIJC mail server.
2. The email resides in the mail server until Bob retrieves it.
3. When Bob uses his laptop to retrieve his emails, the mail server downloads all Bob's mails via the switch to Bob's laptop.
4. Bob can configure the mail server to either delete his mails from the mail server after retrieving or keep them in the mail server so that he can retrieve them again using another client station in the network.

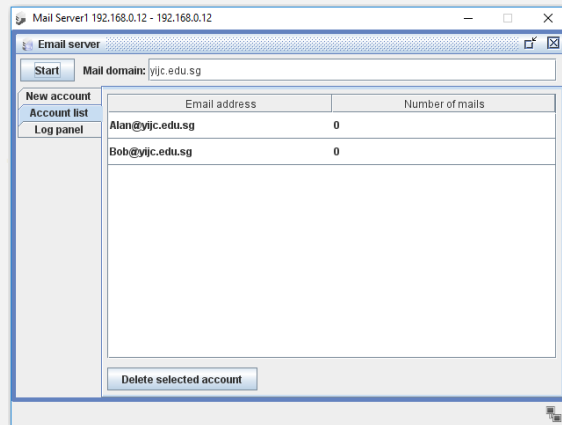


Installing a Mail Server

1 Install the Email Server Software on the Server.

2 Configure the Mail Domain as `yijc.edu.sg`

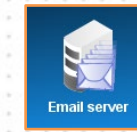
3 Create two new mail accounts for Alan and Bob



4 Click  to start the mail server.

1. Install and configure Mail Server.
2. Configure Mail Exchange in DNS Server.
3. Install and configure Email program in client Notebooks.
4. Test the Mail service.

LAN1 (192.168.0)



Mail Domain:
`yijc.edu.sg`

Mail Server1 192.168.0.12

Switch1 192.168.0

Alan 192.168.0.10

Bob 192.168.0.11

DNS 192.168.2.10

Router

LAN2 (192.168.1)

Switch2 192.168.1

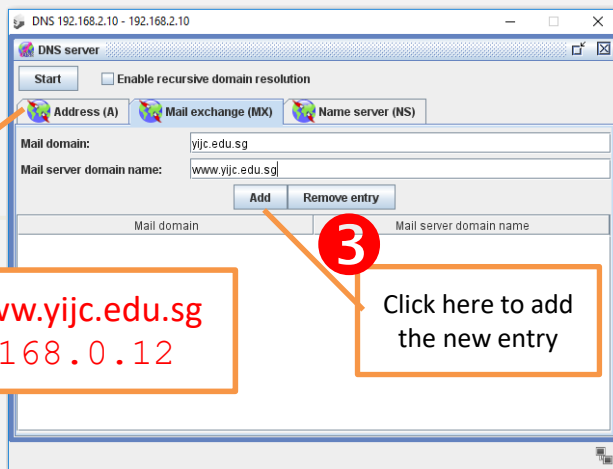
Charles 192.168.1.10

David 192.168.1.11

Installing a Mail Server

1 Select the Mail exchange (MX) tab in the DNS server.

2 Add a new entry with the Mail domain as `yijc.edu.sg` and the Mail server domain name as `www.yijc.edu.sg`

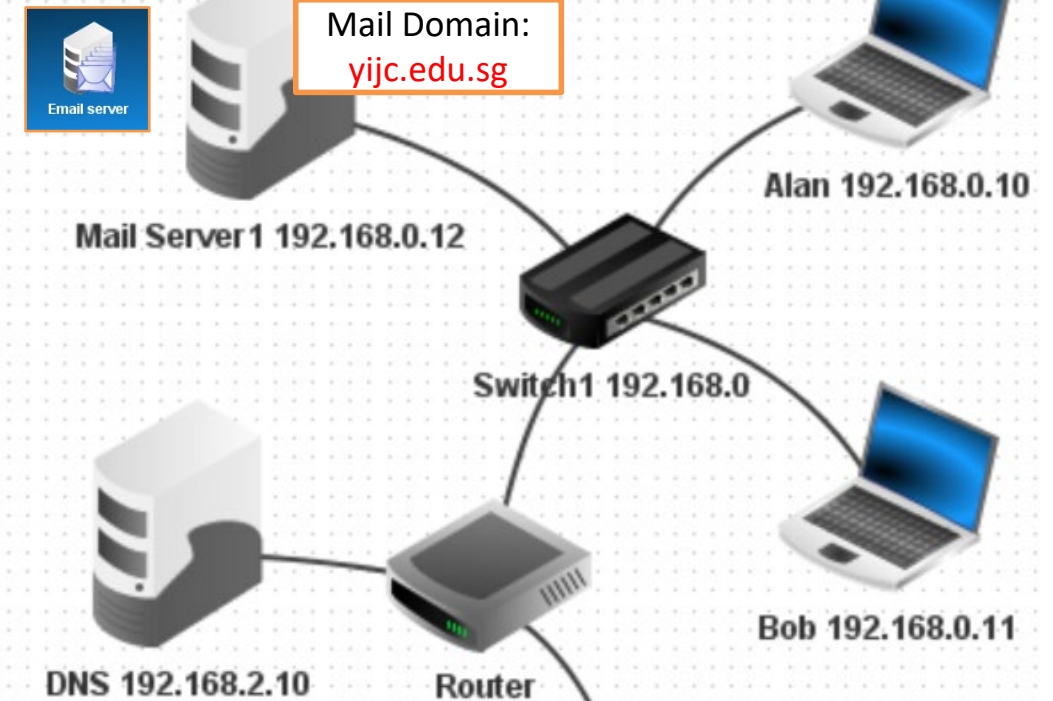


Domain name: `www.yijc.edu.sg`
IP address: `192.168.0.12`

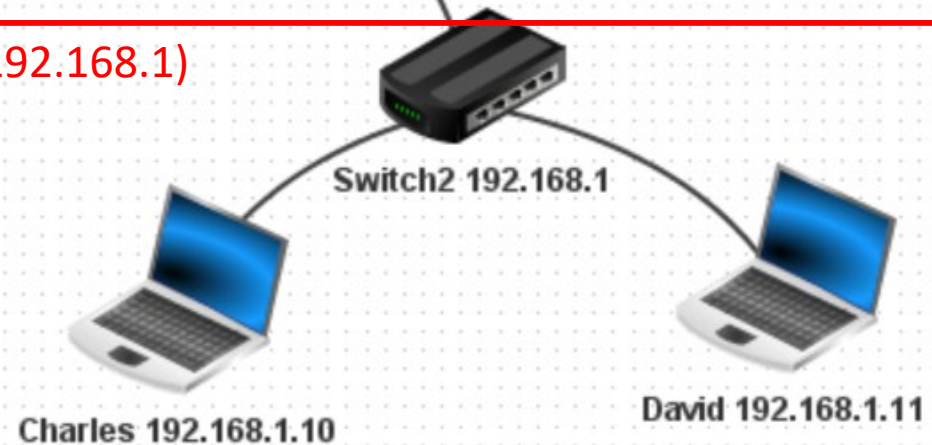
4 Click **Start** to start the DNS server.

1. Install and configure Mail Server.
2. Configure Mail Exchange in DNS Server.
3. Install and configure Email program in client Notebooks.
4. Test the Mail service.

LAN1 (192.168.0)

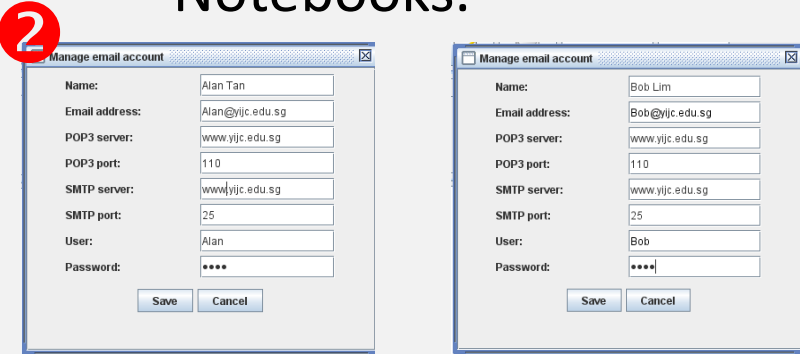


LAN2 (192.168.1)

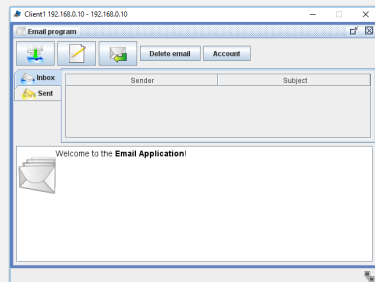


Installing a Mail Server

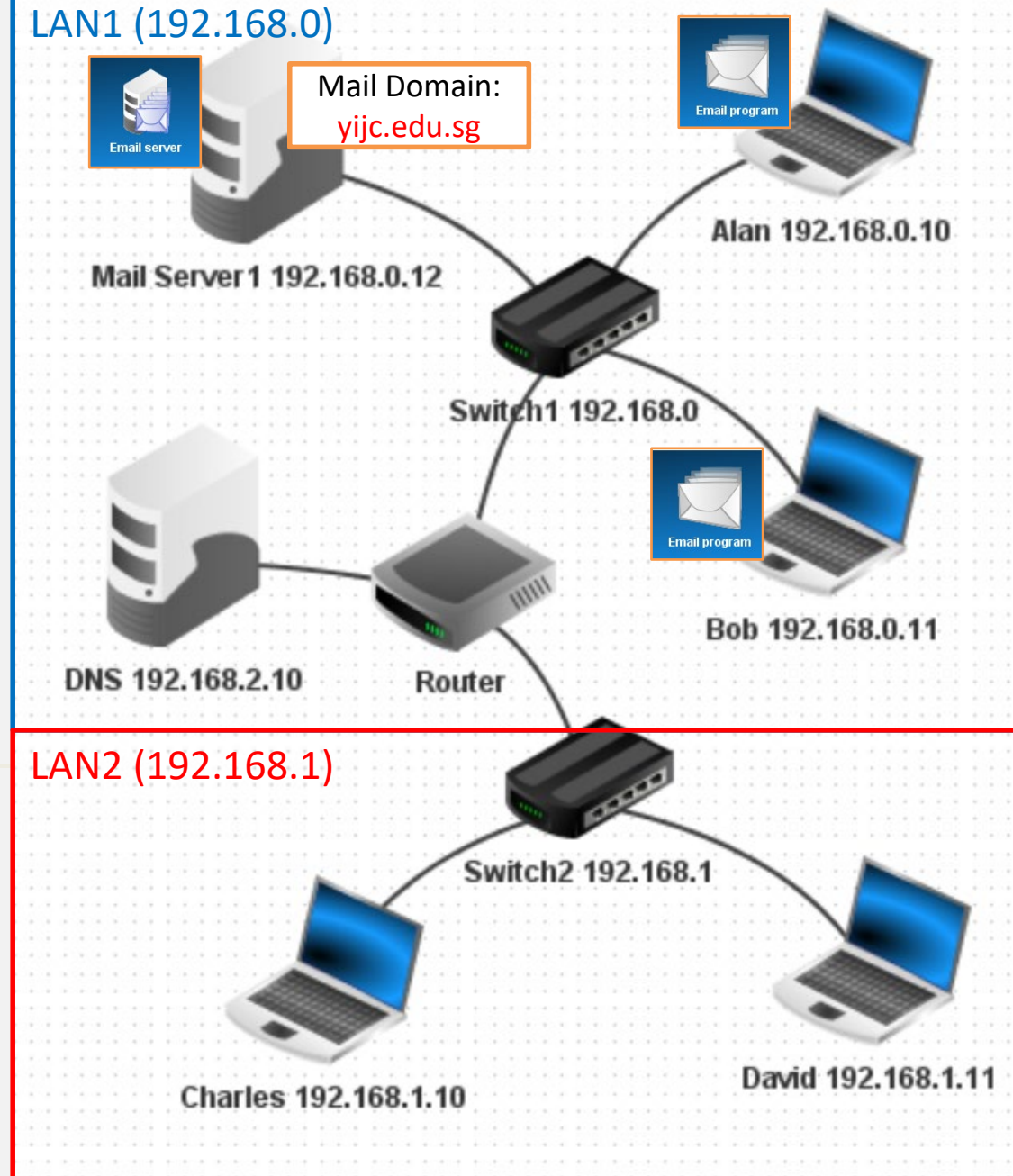
1 Install the Email program on Alan's and Bob's Notebooks.

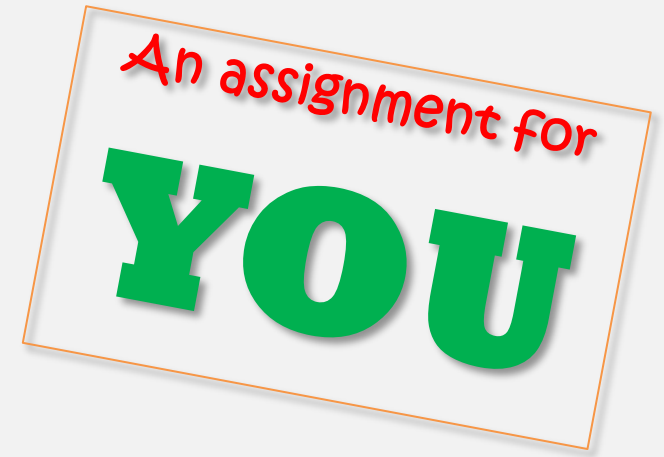


3 Test the mail service by sending emails between Alan and Bob. Observe the **green lines** simulating the transmission of the data.



1. Install and configure Mail Server.
2. Configure Mail Exchange in DNS Server.
3. Install and configure Email program in client Notebooks.
4. Test the Mail service.





Hands-on 7 :

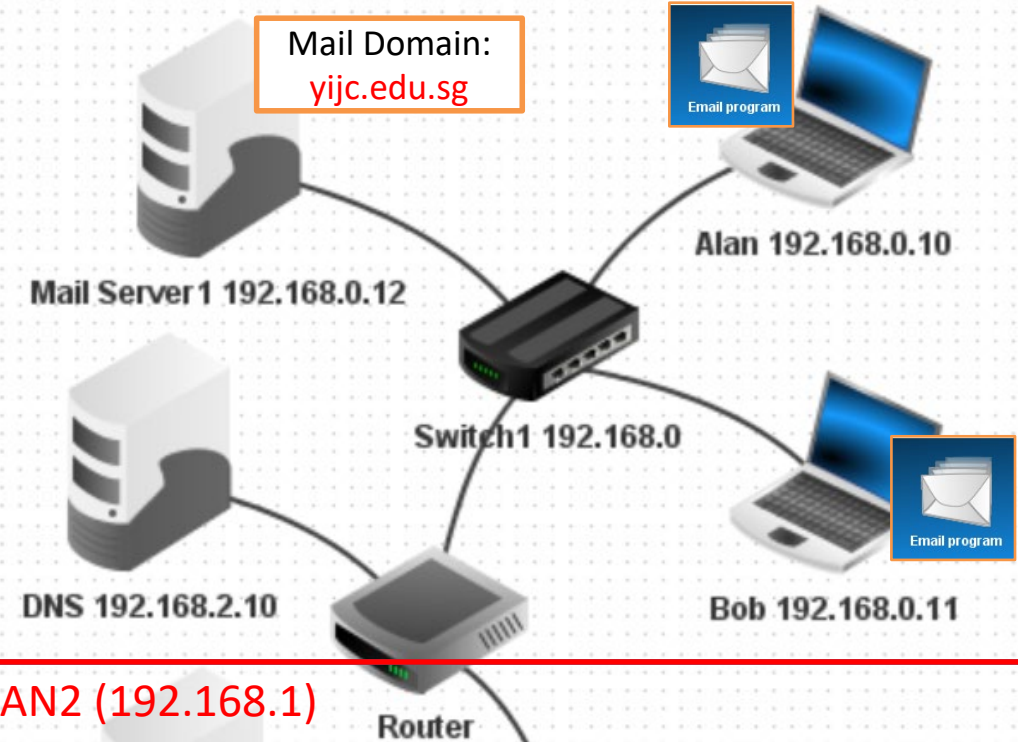
Try-It : Installing another Mail Server

How two Mail Servers work?

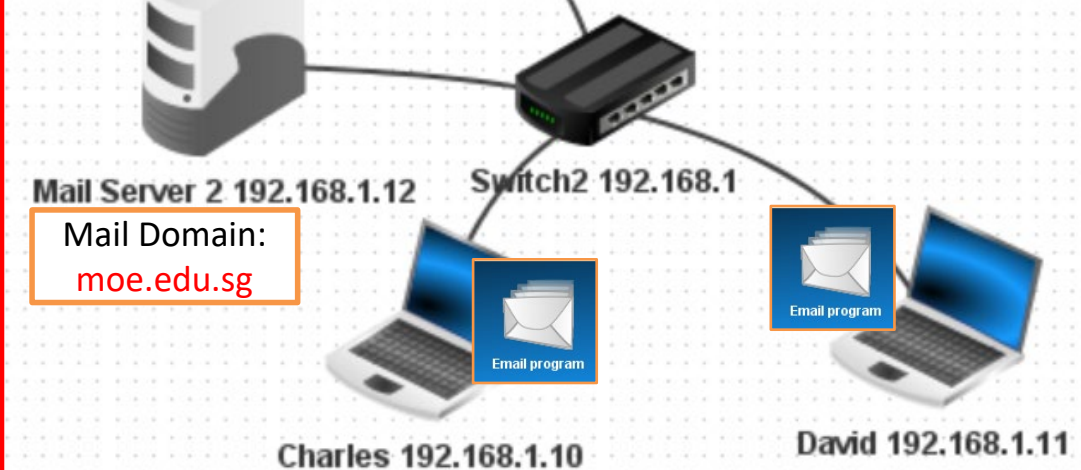
If Alan (alan@yijc.edu.sg) sends an email to David (david@moe.edu.sg):

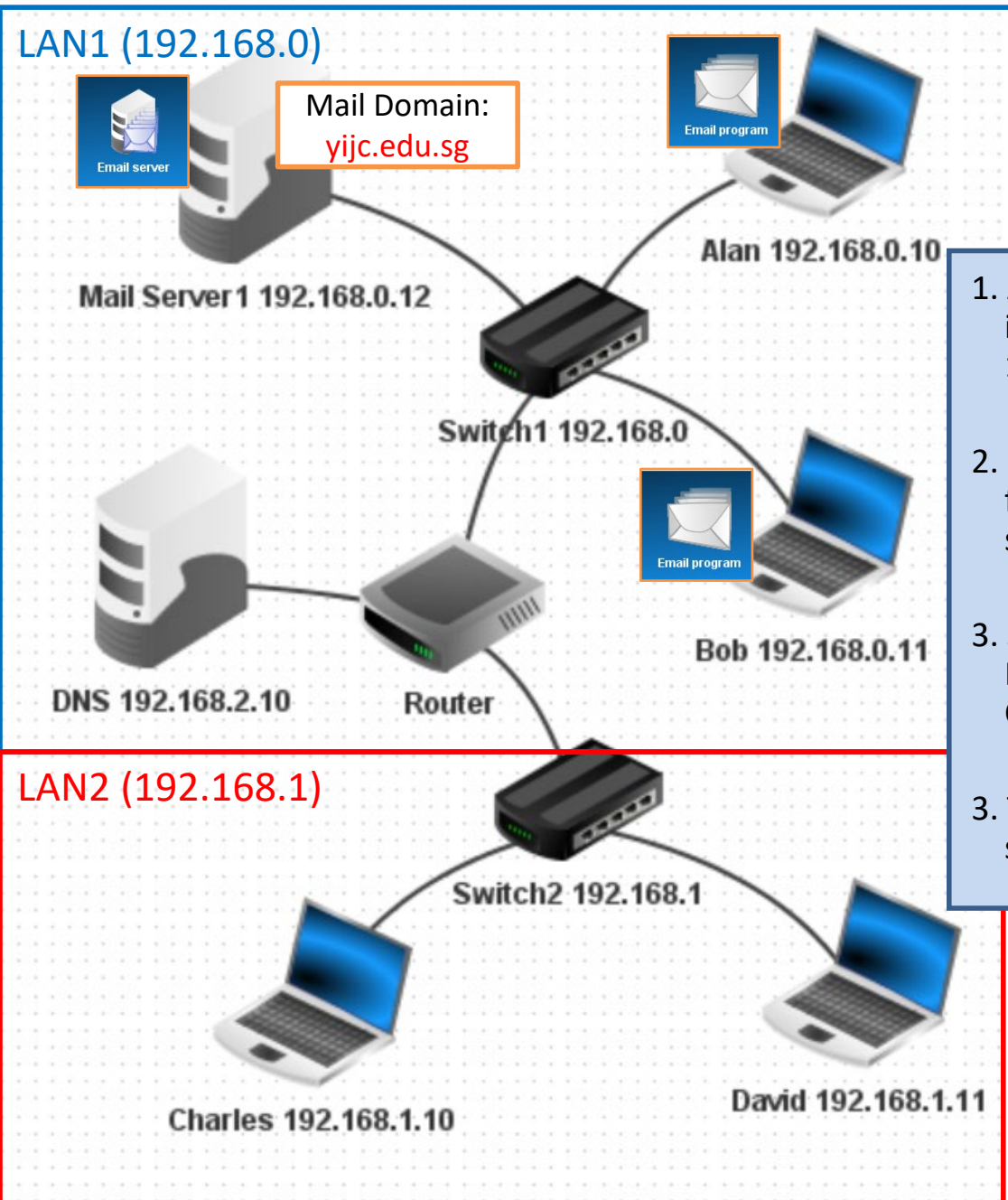
1. The email goes from Alan's laptop to the switch1 and then to the YIJC mail server.
2. The email is then sent via the Switch1 and Router, then via Switch1 to the MOE mail server in the other LAN.
3. This email resides in the MOE mail server until David retrieves it.
3. When David uses his laptop to retrieve his emails, the MOE mail server downloads all David's mails via the switch2 to David's laptop.
4. David can reply Alan's email and it will be delivered in the reverse path.

LAN1 (192.168.0)

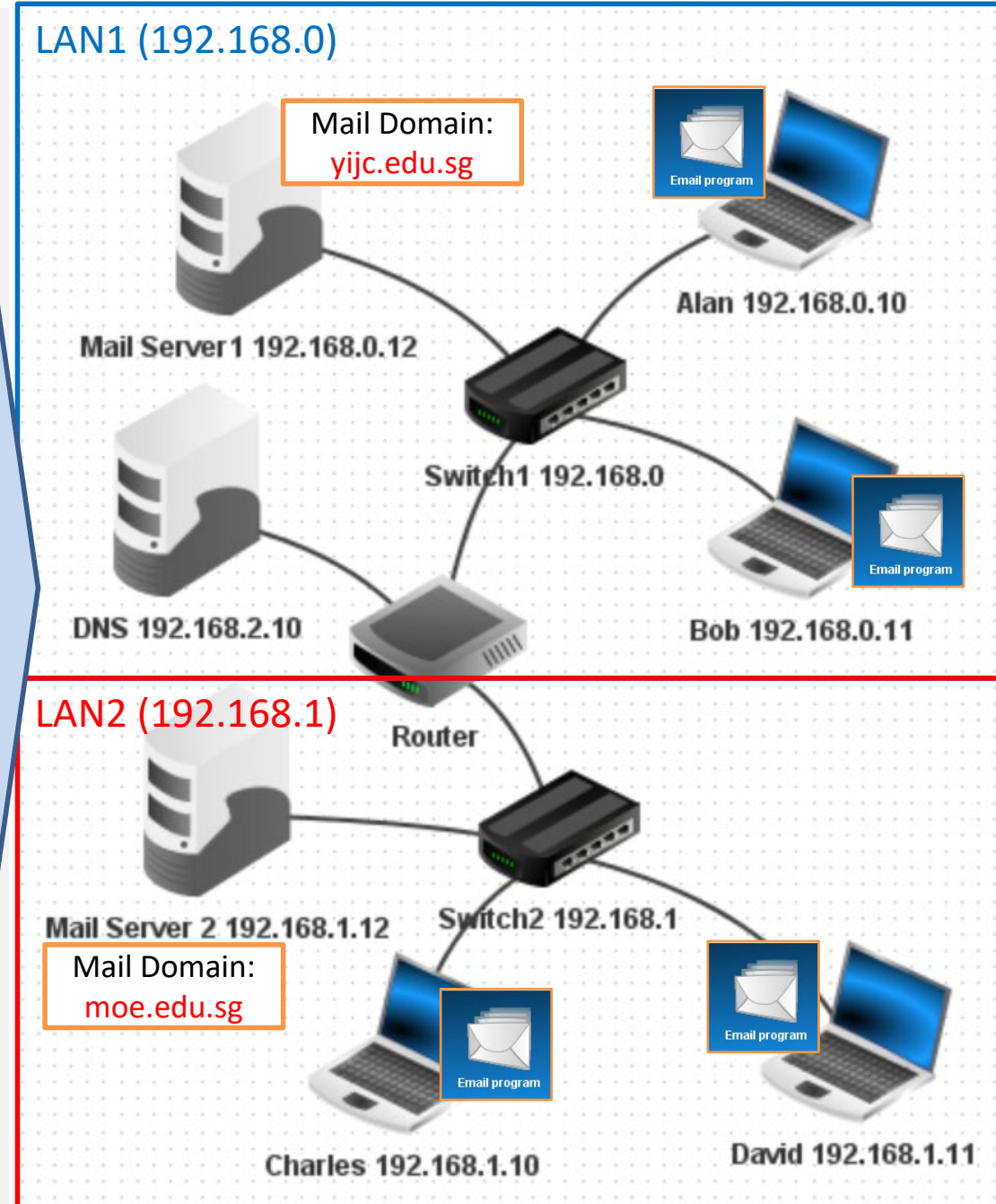


LAN2 (192.168.1)





1. Add a mail server in the LAN 192.168.1
2. Configure the DNS for the new mail server.
3. Setup two client Notebooks for Charles and David
3. Test the mail services



The End

Slide 12 – Circuit Switching vs Packet Switching

<https://www.youtube.com/watch?v=B1tEYnFqL8>

This video covers two main types of switching: circuit switching and packet switching. In circuit switching, a dedicated channel is established for the duration of communication, and it is commonly used in traditional telephone networks. Packet switching, on the other hand, breaks data into packets and sends them through different paths on the network, making it more flexible and efficient. In a traditional telephone network, a call is routed through central offices, and a dedicated channel is established between two parties for the duration of the call. This method can be expensive for long-distance or international calls. In contrast, packet switching networks use routers to send packets of data through different paths to their destination, which is more cost-effective. Packet switching is the basis of the internet and is used for digital data transmission. Circuit switching is still used for traditional telephone calls, but the emergence of voice over IP (VoIP) technology has enabled telephone service over the internet, which is a packet-switched network. In conclusion, circuit switching and packet switching are two different methods of communication in networks. While circuit switching is ideal for real-time communication, packet switching is more flexible and efficient. The emergence of VoIP technology has enabled telephone service over the internet, which is a packet-switched network.

Slide 13 – Secret of Subnet Mask

<https://www.youtube.com/watch?v=yLeuGOOrUvo>

This video explains the subnet mask and how it is used in computer networking. Every computer on a network has an IP address and a subnet mask. The subnet mask is a 32-bit number that is used to determine whether a computer is on the same network as another computer or on a different network. It shows how to find the subnet mask for Mac and Windows computers. The subnet mask is shown in three formats: binary, decimal, and shorthand. The subnet mask has two parts: the ones part and the zeros part. The ones part represents the network ID, and the zeros part represents the host ID. The video gives an example of how a computer uses the subnet mask to determine if another computer is on the same network or on a different network. The subnet mask is used by a computer to determine if any computer is on the same network or on a different network. Networking devices use different pieces of information to communicate locally and remotely.

Slide 20 – Undersea Cables

<https://youtu.be/IIAJJI-qG2k>

NO TRANSCRIPT: Video with no speech.

Video Description:

Animation with words/facts displayed.

Every time you visit a web page or send an email, data is being sent and received through an intricate cable system that stretches around the globe. Since the 1850s, we've been laying cables across oceans to become better connected. Today, there are hundreds of thousands of miles of fiber optic cables constantly transmitting data between nations.

Slide 20 – Network Types

https://youtu.be/4_zSIXb7tLQ

This video explains different types of networks based on their geographical coverage. The first type is a Personal Area Network (PAN), which is a small network used for connecting personal devices such as smartphones, tablets, and laptops, typically using wireless technologies like Bluetooth, infrared, or NFC. Local Area Networks (LAN) are groups of devices located in the same building that use wired or wireless communication, and the most common type of LAN is the Ethernet LAN. Campus Area Networks (CAN) join two or more LANs together within a limited area, such as a university or a corporate campus. Metropolitan Area Networks (MAN) span over several buildings in a city or town, typically using a high-speed connection such as fiber optic cable. Storage Area Networks (SAN) are dedicated high-speed networks used for data storage, while Wide Area Networks (WAN) are the largest type of network that can span over a large geographical area such as a country, continent, or the entire globe. An example of a WAN is the internet. The video gives examples and explains how each network type is used.

Slide 22 – What is MAC address?

<https://youtu.be/UrG7RTWIJak>

Each device connected to a network has a media access control (MAC) address that identifies the device on the local network, while Internet Protocol (IP) addresses are used to identify a device on the internet. When data arrives from the internet to the router, it needs to decide which device to send that data to, and it does this by assigning a private IP address, usually starting with 192.168, to each device and tracking their MAC addresses. The router attaches the correct private IP address to data packets so that they can be sent to the correct device. The MAC address on data packets is constantly changed as the data passes through numerous servers and routers before finally arriving at the destination device. The MAC address is essential for determining where the data packets should go. The MAC addresses are normally hardcoded into the firmware of the device, making it easy to spot a malfunctioning device on the network, and they are used to restrict access to only known devices. It is possible to spoof a MAC address for privacy reasons, but it may not make the internet service provider happy.

Slide 23 – 7 LAN Topologies

<https://youtu.be/T5xQJk-Aamk>

The topic discussed is LAN topology, which refers to how network devices are connected in the physical layer of the OSI model. There are three categories of topology: physical, logical, and hybrid physical. Physical topology is how devices are physically connected, while logical topology is how devices are connected in regards to signal flow. Hybrid topology is a combination of physical and logical topology. There are eight different topologies, including point-to-point, point-to-multipoint, bus, ring, star, mesh, and hybrid topologies. Each has its own benefits and drawbacks, such as fault tolerance, scalability, maintenance, and security. Fully meshed topology means every node connects to every other node, while partially meshed topology is similar but with fewer connections. The formula to calculate the number of connections in fully meshed topology is $C = n(n-1)/2$, where C is the number of connections and n is the number of devices. Fully meshed topology can be expensive and impractical for a large number of devices.

Slide 26 – How a switch builds Source Address Table dynamically?

<https://youtu.be/DukuFSZH9Qw>

In this video, the speaker discusses how a layer 2 switch dynamically builds a source address table (SAT) to determine where to forward traffic on a switched LAN. The SAT is initially built when the switch is turned on by copying the source MAC addresses and building a table of connected devices and their associated ports. The switch updates the SAT dynamically when it receives a frame from a new device. If a destination MAC address is not in the SAT, the switch broadcasts the frame to all other ports except the source port, and the device with the matching MAC address responds with its location. The switch updates the SAT with the new device's MAC address and associated port.

A switch separates traffic, giving each conversation full bandwidth of the network. A switch creates a broadcast domain, where any node using a broadcast address is heard by all other nodes in the same domain. The switch forwards frames based on MAC addresses and uses the SAT to determine where to forward traffic on a switched LAN.

Slide 30 – Hub, Switch, & Router Explained - What's the difference?

https://youtu.be/1z0ULvg_pW8

The article explains the difference between a hub, switch, and router in networking. A hub is used to connect network devices together but is not intelligent and will rebroadcast data to every port, leading to security concerns and unnecessary traffic. A switch is similar to a hub but is intelligent and will direct data packets only to the intended destination port based on the MAC address of the device. A router, on the other hand, routes or forwards data from one network to another based on IP addresses. It inspects the data's IP address and determines if the packet was meant for its own network or if it's meant for another network. If the data packet is not meant for its own network, the router sends it off to another network. Hubs and switches are used to exchange data within a local area network, while routers are used to exchange data between networks, such as out on the internet. The article concludes that hubs and switches create networks, while routers connect networks.

Slide 35 – Why both IP and MAC?

<https://youtu.be/oGoWqdlAOMI>

The use of both IP and MAC addresses in communication is necessary in the TCP/IP world. MAC addresses are unique IDs for devices, and they efficiently facilitate communication within small networks as each device can broadcast to find and communicate with its physical address. On the other hand, IP addresses are like mailing addresses that tell where to find a device on the internet, allowing for mobility as they can change when a device moves to a different network. If we only used MAC addresses, communication would be impossible as everyone would broadcast to find other computers, and if something happened to the hardware NIC card, existing connections would be lost. If we only used IP addresses, we would not know who we are communicating with as they only indicate where a device is. Therefore, both IP and MAC addresses are necessary to ensure that messages are delivered to the correct device on the correct network.

Slide 39 – What is a Server?

<https://youtu.be/UjCDWCeHCzY>

A server is a dedicated computer that provides services to clients, and it can handle a specific service or multiple services on one machine. The type of server setup that an organization chooses depends on its needs. Even though servers are usually powerful centralized computers, any ordinary desktop computer can serve as a server. Desktop computers have limitations in handling large workloads and can't handle a lot of incoming connections from users. Servers need to be reliable and run 24/7 to prevent disruptions to the organization. They require robust hardware and software, and servers must be up and running at all times. Xeon processors are suitable for servers because they are fast and can perform multiple tasks simultaneously. They also support ECC RAM, which is error code correcting memory, and a larger amount of RAM. In addition, servers should have hot swappable hard drives in a RAID configuration, redundant power supplies, and use a server operating system, such as Linux, Windows Server, or mac OS server, to handle thousands of concurrent connections. There are various types of servers, including web servers, email servers, and file servers.

Slide 43 – How does the internet works?

https://youtu.be/7_LPdttKXPc

In simple terms, the internet is a wire (fiber optics, copper, or satellite) that connects computers directly. A server is a special computer connected directly to the internet that hosts web pages or files. Every server has a unique Internet Protocol address or IP address, like a postal address, that helps computers find each other. Computers that are not directly connected to the internet are called clients and are connected through an internet service provider (ISP). Information transmitted through the internet, such as web pages, emails, and pictures, are broken down into smaller pieces called packets. Routers direct these packets around the internet, helping each packet get closer to its destination. Each packet has an IP address that identifies where it should go and routers unwrap layers of identification to direct the packet to its destination. IP addresses and routers prevent packets from ending up in the wrong place. Overall, understanding the basics of the internet, such as its wire-based structure, client-server architecture, packet transmission, and routing mechanisms, can help users become more tech-savvy and informed internet users.

Slide 43 – The difference between the Internet and the Web

<https://youtu.be/UVkT59PmRdo>

The terms "internet" and "web" are often used interchangeably, but they actually have different meanings. The internet is the network of machinery that allows access to the web, which is a collection of pages with text and images. The web is a subset of internet traffic, much like cars on a physical highway. Other types of internet traffic include email and media. Therefore, the internet contains the web, but the web is not the same as the internet.

Slide 45 – How does the Internet works?

<https://youtu.be/x3c1ih2NJEg>

This video explains how the internet works. The journey of data from a data center to a user's device is described. The video states that the use of satellites to transmit data is not efficient because of latency caused by the distance the data would have to travel. Instead, data is transmitted using a complex network of optical fiber cables, which connect the data center to the user's device. Devices connected to the internet are identified by a unique IP address, which serves as the address through which information is sent. Domain names, such as YouTube.com, are used instead of IP addresses, making it easier for users to remember. DNS is a large phone book that translates domain names to IP addresses. Once the server gets a request to access a website, the data flow starts, and data is transferred via optical fiber cables in the form of light pulses. The video also explains how optical fiber cables are laid and maintained, and how they are connected to a router that converts light signals to electrical signals. Finally, the video explains how cellular data is used to access the internet.

Slide 49 – How the Domain Name Server (DNS) works?

<https://youtu.be/mpQZVYPuDGU>

In networking, computers communicate using numbers such as IP addresses, while humans use names to identify things. To bridge this communication gap, DNS was developed, which resolves domain names to IP addresses. When a user types in a web address in their browser, the DNS server searches through its database to find the matching IP address for that domain name. If it can't find it, it will send the query to the resolver server, which is usually the user's Internet service provider. If the resolver can't find the IP address, it will send the query to the root server, which directs it to the TLD (top level domain) server for that domain. The TLD server will then direct the query to the authoritative name server, which holds the IP address for that domain. Once the authoritative name server responds with the IP address, the resolver stores it in its cache memory so it can retrieve it quickly if needed again in the future. DNS works like a phone book, where you look up the name first, and it gives you the number.

Slide 56 – DHCP Explained – Dynamic Host Configuration Protocol

<https://youtu.be/e6-TaH5bkjo>

In the video, the concept of DHCP or Dynamic Host Configuration Protocol is discussed. DHCP is a protocol that automatically assigns IP addresses to computers on a network. This process makes managing a network much easier than the original method of manually assigning an IP address to each computer. With DHCP, a server assigns IP addresses from a range called a scope, which can be customized by the network administrator. The server assigns IP addresses as a lease, which is a specific amount of time that the computer can use that IP address. This helps to ensure that the server does not run out of IP addresses in its scope. If a computer is removed from the network, the IP address it was using is returned to the DHCP server and can be used for another computer. The video explains that a dynamic IP is the best choice for assigning an IP address because it is automatic and saves time. Finally, the video demonstrates how to verify the DHCP settings by opening a command prompt on a Windows computer and typing in `ipconfig /all`.

Slide 56 – DHCP Explained – Dynamic Host Configuration Protocol

<https://youtu.be/e6-TaH5bkjo>

In the video, the concept of DHCP or Dynamic Host Configuration Protocol is discussed. DHCP is a protocol that automatically assigns IP addresses to computers on a network. This process makes managing a network much easier than the original method of manually assigning an IP address to each computer. With DHCP, a server assigns IP addresses from a range called a scope, which can be customized by the network administrator. The server assigns IP addresses as a lease, which is a specific amount of time that the computer can use that IP address. This helps to ensure that the server does not run out of IP addresses in its scope. If a computer is removed from the network, the IP address it was using is returned to the DHCP server and can be used for another computer. The video explains that a dynamic IP is the best choice for assigning an IP address because it is automatic and saves time. Finally, the video demonstrates how to verify the DHCP settings by opening a command prompt on a Windows computer and typing in `ipconfig /all`.

Slide 58 – What is SMTP - Simple Mail Transfer Protocol

<https://www.youtube.com/watch?v=PJo5yOtu7o8>

In this video, it introduces the SMTP, which stands for Simple Mail Transfer Protocol, the protocol used for sending emails. SMTP is a set of commands that authenticate and direct the transfer of email, ensuring that it reaches its destination, assuming the recipient's email address is correct and still exists. SMTP uses the TCP protocol, which guarantees the delivery of the email.

SMTP is configured in email clients such as Microsoft Outlook or mobile devices, and if using Gmail, the SMTP server address would be smtp.gmail.com. When an email is sent from an email client, it travels to the SMTP server, which sends the message to the recipient's email server, also using SMTP. The email then stays on the recipient's email server until the recipient logs into their email account and downloads the email using POP or IMAP or views it on the server using webmail.

The video suggests associating SMTP with "Sending Mail To People" to remember what it does, and compares it to a mailman who picks up mail and delivers it to the destination mailbox.